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Systems and methods for an autonomous mobile robot

Abstract

An autonomous robotic cart includes a chassis, sensors coupled with the chassis, visible light cameras, and a handlebar unit coupled with the chassis. The handlebar unit includes a handlebar and a force sensor configured to detect a translational force and a rotational force exerted on the handlebar. The autonomous robotic cart also includes a holonomic and omnidirectional mechanical drive unit coupled with the chassis. The autonomous robotic cart is configured to autonomously navigate a physical environment to execute one or more navigation goals determined based on communication with a remote computing system configured to manage a fleet of robots including the autonomous robotic cart and also to cause the autonomous robotic cart to move translationally and rotationally in a direction corresponding to a output force vector determined based on sensor data.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to of U.S. Provisional Patent Application 63/560,381 filed on Mar. 1, 2024 by Jules et al., titled “AUTONOMOUS MOBILE ROBOT”, and to U.S. patent application Ser. No. 18/655,609, filed May 6, 2024 by Luong et al., titled “Autonomous Robot with Force Sensing User Handlebar”, which claims priority to U.S. Provisional Patent Application 63/571,352 by Luong et al., titled: “Autonomous Robot with Force Sensing User Handlebar”, filed on Mar. 28, 2024, all of which are incorporated herein by reference in their entirety and for all purposes.

FIELD OF TECHNOLOGY

(1) This patent application relates generally to robotics, and more specifically to autonomous mobile robots.

BACKGROUND

(2) Autonomous and semi-autonomous robots can be operated without user input, but in some situations user operation is desirable. User operation of autonomous and semi-autonomous robots may be achieved through a control interface, such as a graphical user interface to allow a user to control the autonomous or semi-autonomous robot from the point of view of the robot. However, autonomous and semi-autonomous robots may operate in environments with other human workers. Such workers may not be specifically tasked with operating the autonomous and semi-autonomous robots and may not include specific devices to do so, but may still find themselves in situations where they may need to operate such autonomous and semi-autonomous robots. Accordingly, improved mechanisms and techniques for autonomous robots capable of being controlled by humans are desired.

SUMMARY

(3) Techniques and mechanisms described herein provide systems, methods, and non-transitory computer readable media having instructions stored thereon for an autonomous robotic cart.

According to various embodiments, the autonomous robotic cart includes a chassis including one or more shelves and a plurality of sensors coupled with the chassis. The plurality of sensors includes a plurality of visible light cameras. The autonomous robotic cart also includes a handlebar unit coupled with the chassis and including a handlebar and a force sensor configured to detect a translational force and a rotational force exerted on the handlebar. The autonomous robotic cart also includes a holonomic and omnidirectional mechanical drive unit coupled with the chassis and a processor.

(4) According to various embodiments, the processor is configured to transmit a first one or more instructions to the holonomic and omnidirectional mechanical drive unit to autonomously navigate a physical environment to execute one or more navigation goals determined based on communication with a remote computing system configured to manage a fleet of robots including the autonomous robotic cart. The processor is also configured to determine an output force vector based on the translational force, the rotational force, and sensor data received from the plurality of sensors, and to transmit a second one or more instructions to the holonomic and omnidirectional mechanical drive unit to cause the autonomous robotic cart to move translationally and rotationally in a direction corresponding to the output force vector. The autonomous robotic cart also includes a communication channel communicatively coupling the processor with the holonomic and omnidirectional mechanical drive unit, the plurality of visible light cameras, and the force sensor.

(5) According to various embodiments, the processor is configured to update a scene graph representing the physical environment based at least in part on the sensor data. The output force vector may be determined based at least in part on an obstacle avoidance vector representing a virtual force on the autonomous robotic cart from a designated direction corresponding with an obstacle represented in the scene graph. The obstacle avoidance vector may have a magnitude that increases in inverse proximity of the autonomous robotic cart with the obstacle. The scene graph may be updated based at least in part on the communication with the remote computing system.

(6) In some embodiments, the processor may detect a human within the physical environment based on the sensor data. The output force vector may be determined based at least in part on a human avoidance vector representing a virtual force on the autonomous robotic cart from a designated direction corresponding with the human.

(7) In some implementations, the autonomous robotic cart is backdriveable. The processor may be configured to transmit an instruction to the holonomic and omnidirectional mechanical drive unit to negate a force exerted on the chassis of the autonomous robotic cart.

(8) In some embodiments, the holonomic and omnidirectional mechanical drive unit is configured to cause the autonomous robotic cart to move along a planar surface, and the handlebar is positioned substantially orthogonal to the planar surface.

(9) In some implementations, the shelves may be positioned in two or more layers. The chassis may include one or more vertical support elements supporting the layers. The handlebar may be incorporated into a vertical support element of the one or more vertical support elements.

(10) In some embodiments, the force sensor may detect the translational force and the rotational force based on a potential difference across an electrical conductor that is transverse to an electric current in the electrical conductor and to an applied magnetic field perpendicular to the electric current.

(11) In some embodiments, the holonomic and omnidirectional mechanical drive unit includes four powered wheels. Alternatively, or additionally, the holonomic and omnidirectional mechanical drive unit includes four unpowered wheels. Alternatively, or additionally, the holonomic and omnidirectional mechanical drive unit includes two powered bidirectional wheels coupled with a turntable rotating about an axis. An unpowered freely rotating caster wheel may be coupled with the turntable.

(12) In some implementations, the autonomous robotic cart includes a light component configured to illuminate a region of the one or more shelves corresponding to a current or intended location of

an object designated for placement retrieval. Alternatively, or additionally, the autonomous robotic cart includes a light component may configured to illuminate a region of the physical environment corresponding to a current or intended location of an object designated for placement or retrieval. (13) In some embodiments, autonomously navigating the physical environment may involve moving along a nominal trajectory that includes one or more waypoints while avoiding objects encountered along the nominal trajectory. (14) The objects may include one or more humans moving throughout the physical environment, and the autonomous robotic cart may be configured to identify the one or more humans as humans based on sensor data received from the plurality of sensors. Alternatively, or additionally, the objects may include one or more autonomous mobile robots moving throughout the physical environment, and the autonomous mobile robot may be configured to avoid the autonomous mobile robots based at least in part on location information received from the remote computing system.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The included drawings are for illustrative purposes and serve only to provide examples of possible structures and operations for the disclosed inventive systems, apparatus, methods, and computer program products for an autonomous mobile robot. These drawings in no way limit any changes in form and detail that may be made by one skilled in the art without departing from the spirit and scope of the disclosed implementations.
- (2) FIG. 1 illustrates a perspective view of an autonomous robot, configured in accordance with one or more embodiments.
- (3) FIG. 2 illustrates a perspective view of an autonomous robot with a horizontal handlebar, configured in accordance with one or more embodiments.
- (4) FIG. 3 illustrates an additional configuration of an autonomous mobile robot, arranged in accordance with one or more embodiments.
- (5) FIG. 4 illustrates a disassembled view of an autonomous mobile robot, configured in accordance with one or more embodiments.
- (6) FIG. 5 illustrates a communication architecture diagram of the autonomous mobile robot, configured in accordance with one or more embodiments.
- (7) FIG. 6 is a block diagram of a computing device, configured in accordance with one or more embodiments.
- (8) FIG. 7 illustrates a diagram of a drive unit, configured in accordance with one or more embodiments.
- (9) FIG. 8 illustrates a view of an autonomous mobile robot that includes a drive assembly containing two drive units, configured in accordance with one or more embodiments.
- (10) FIG. 9A, FIG. 9B, and FIG. 9C illustrates views of an autonomous mobile robot that includes a drive assembly containing various numbers of drive units, configured in accordance with one or more embodiments.
- (11) FIG. 10A, FIG. 10B, and FIG. 10C illustrate perspective views of portions of an autonomous mobile robot, configured in accordance with one or more embodiments.
- (12) FIG. 11A and FIG. 11B illustrate handlebars for guiding an autonomous mobile robot, configured in accordance with one or more embodiments.
- (13) FIG. 12 illustrates a force sensor, configured in accordance with one or more embodiments.
- (14) FIG. 13 illustrates a perspective view of a force sensing handlebar, configured in accordance with one or more embodiments.
- (15) FIG. 14 illustrates a perspective view of portions of a force sensing handlebar, configured in accordance with one or more embodiments.

- (16) FIG. 15 illustrates a top view of a force sensing handlebar for an autonomous robot in a first manipulated position, configured in accordance with one or more embodiments.
- (17) FIG. 16 illustrates a top view of a force sensing handlebar for an autonomous robot in a second manipulated position, configured in accordance with one or more embodiments.
- (18) FIG. 17 illustrates a perspective view of an autonomous robot with a force sensing base, configured in accordance with one or more embodiments.
- (19) FIG. 18 illustrates a perspective view of portions of a force sensing base, configured in accordance with one or more embodiments.
- (20) FIG. 19 illustrates a method of controlling an autonomous mobile robot, performed in accordance with one or more embodiments.
- (21) FIG. 20 illustrates a method for executing a robot control loop, performed in accordance with one or more embodiments.
- (22) FIG. 21 illustrates an architecture diagram for a control portion of a mobile robot, configured in accordance with one or more embodiments.
- (23) FIG. 22 illustrates a method for determining a robot control output instruction, performed in accordance with one or more embodiments.
- (24) FIG. 23 illustrates a method for autonomous motion control of an autonomous mobile robot, performed in accordance with one or more embodiments.
- (25) FIG. 24 illustrates a diagram showing sensor placement on an autonomous mobile robot configured in accordance with one or more embodiments.
- (26) FIG. 25A illustrates a view of an attachment point between a lighting element and a communication channel in an embodiment including an angled shelf configuration.
- (27) FIG. 25B illustrates a view of an attachment point between a lighting element and a communication channel in an embodiment including a flat shelf configuration.
- (28) FIG. 26 illustrates a method for controlling one or more lights associated with an autonomous mobile robot, performed in accordance with one or more embodiments.
- (29) FIG. 27 illustrates a block diagram representation of a shelf that may form part of an autonomous mobile robot, configured in accordance with one or more embodiments.
- (30) FIG. 28A, FIG. 28B, and FIG. 28C illustrate diagrams of situations in which haptic feedback may be provided via a mechanical drive unit, generated in accordance with one or more embodiments.
- (31) FIG. 29A, FIG. 29B, and FIG. 29C illustrate diagrams of situations in which haptic feedback may be provided via a mechanical drive unit, generated in accordance with one or more embodiments.
- (32) FIG. 30A and FIG. 30B illustrate diagrams of situations in which haptic feedback may be provided via a mechanical drive unit, generated in accordance with one or more embodiments.
- (33) FIG. 31 illustrates a method of determining a virtual friction vector, performed in accordance with one or more embodiments.
- (34) FIG. 32 illustrates a method of calibrating a physical friction vector for an autonomous mobile robot, performed in accordance with one or more embodiments.

DETAILED DESCRIPTION

(35) Introduction

(36) Techniques and mechanisms described herein provide for an autonomous mobile robot configured to operate in cooperation with people. In some embodiments, the autonomous mobile robot may be configured as a cart capable of transporting one or more objects. The robot may operate in one of various modes. For example, in an autonomous mode the robot may operate without physical human intervention, for instance autonomously moving from one location to another and/or performing various types of tasks. As another example, in a robot-guided mode, the robot may direct a human to perform a task, such as guiding a human from one location to another. As another example, in a person-guided mode, the robot may operate in a manner responsive to

human guidance. The robot may be configured to seamlessly switch between such modes, for instance with the aid of computer vision, user interaction, and/or artificial intelligence.

(37) In some embodiments, an autonomous mobile robot may be configured for operation in a warehouse environment. For example, the robot may be equipped and configured to perform and support warehouse operations such as item picking, item transport, and item replenishment workflows. As another example, the robot may be equipped to perform automated item pickup and/or dropoff, for instance via one or more arms or conveyer belts. As still another example, the robot may be equipped to perform automated charging and/or battery swapping. As yet another example, the robot may be equipped to autonomously navigate to a particular location, follow a user, respond to user instructions, amplify a force exerted on the robot by a user, and/or perform other types of operations. The robot may be adapted to site-specific environmental conditions and/or processes.

(38) The robot may include a drive assembly to provide motive power. In some embodiments, the drive assembly may include one or more drive units, with each drive unit orientable in an independent manner to that of the other drive units. Each drive unit may include a plurality of driven wheels that may be independently driven. Independent drive of each of the drive wheels of the drive assembly allows for the drive assembly to move the robot in a holonomic manner. That is, the robot may be driven without constraints in direction of motion.

(39) In some embodiments, the robot may include a force sensing assembly to allow for a user to manipulate the robot. The force sensing assembly may include, for example, a handlebar. The handlebar may be mounted in any orientation, such as in a horizontally or vertically mounted orientation. Additionally or alternatively, the force sensing assembly may be a force sensing base or another mechanism configured to receive physical input from a user (e.g., a hand, arm, foot, or leg of a user).

(40) In some implementations, a user may manipulate the force sensing assembly by, for example, providing force to a handlebar to move the handlebar from a neutral position. The manipulation of the force sensing assembly may provide instructions to the robot and cause the drive assembly to move the robot in accordance with the instructions provided via the force sensing assembly. Such commands may, for example, override autonomous or semi-autonomous operation of the robot.

(41) Techniques and mechanisms described herein also provide for control of a robot, which may be one or more of omnidirectional, holonomic, backdrivable, and autonomous. Input may be received from a force sensor identifying a force exerted on the force sensor. Based on this input, a physical input force vector quantifying a force exerted on the force sensor in two or more dimensions may be determined. A force output vector may be determined by combining the physical input force vector with a second force input vector. The force output vector may quantify a force to apply to move the robot in another direction. The second force input vector may include, for instance, a frictional force and/or a functional force determined based on one or more operational objectives. The force output vector may include a force multiplier multiplying the physical force exerted on the force sensor. An indication of the force output vector may be sent to a mechanical drive unit at the robot and then used to direct the movement of the robot via the mechanical drive unit.

(42) In some embodiments, an autonomous mobile robot may support omnidirectional movement. That is, the autonomous mobile robot may be capable of movement in any direction.

(43) In some embodiments, an autonomous mobile robot may support holonomic movement. That is, the autonomous mobile robot may be capable of powered movement in any direction corresponding with a degree of freedom associated with the robot. For instance, a conventional automobile is not holonomic because it has three motion degrees of freedom (i.e., x, y, and orientation) but only two controllable degrees of freedom (i.e., speed and steer angle). In contrast, a conventional train is holonomic because it has one controllable degree of freedom (i.e., speed) and one motion degree of freedom (i.e., position along the track).

(44) In some embodiments, an autonomous mobile robot may support omnidirectional and holonomic movement. That is, the autonomous mobile robot may be capable of powered movement and rotation in any direction from any position.

(45) In some embodiments, an autonomous mobile robot may be backdriveable. That is, the drive unit may operate so as to maintain a desired force of interaction at a level close to zero. For instance, when pressure is exerted on the robot, even in an area other than the handlebar, the drive unit may operate to move the robot in a direction consistent with the force so as to reduce the force of interaction. For example, if a person were to exert 10 Newtons of force on a fixed object, such as a wall, the wall would exert an equal and opposite force on the person due to the wall's immobility, causing the person to experience 10 Newtons of force in the opposite direction of the force. In contrast, if a person were to exert 10 Newtons of force on a backdriveable autonomous mobile robot, the drive unit may cause the robot to move in the direction of the force at a speed such that the person would experience approximately 0 Newtons of force. The robot may be configured to control the drive unit to keep the level of force experienced by the operator below a designated threshold under normal operating conditions.

(46) According to various embodiments, an autonomous mobile robot may synthesize various types of instructions and input to provide a seamless experience. In some configurations, an autonomous mobile robot may support one or more of the following mobility input mechanisms. First, an autonomous mobile robot may be backdriveable in the sense that the drive unit may operate to move in a direction to effectively cancel out force exerted on the robot from any direction. Second, an autonomous mobile robot may be responsive to force exerted on a force sensor, for instance by instructing the drive unit so as to multiply the exerted force in the direction of movement. Third, an autonomous mobile robot may respond to the presence of a human, for instance based on touch sensor and/or optical sensor data. As one example, a robot may cease autonomous movement and wait for more instructions when grasped or approached by a human. Fourth, an autonomous mobile robot may autonomously navigate along a nominal trajectory based on information determined at the autonomous mobile robot and/or information received from a remote computing device, such as a fleet controller. Fifth, an autonomous mobile robot may autonomously act in support of operational guidelines and objectives, for instance to avoid both static obstacles (such as walls) and dynamic obstacles (such as humans).

(47) When using conventional techniques and mechanisms, onboarding autonomous mobile robots in an industrial setting takes a significant amount of time. In contrast, various embodiments described herein facilitate rapid onboarding. In some embodiments, an autonomous mobile robot can be on-boarded without bringing an autonomous mobile robot on-site for an initial survey. Such rapid deployment can significantly increase adoption speed.

(48) When using conventional techniques and mechanisms, even small changes to autonomous mobile robot configuration and workflows cannot be made in real time. In contrast, various embodiments described herein provide for easy adjustments to daily workflows without intervention by a technical support team.

(49) When using conventional techniques and mechanisms, industrial autonomous mobile robots are typically configured with expensive hardware that is customized to particular environments. In contrast, various embodiments described herein provide for autonomous mobile robots may be configured with standardized hardware and software that is easily and cheaply applicable and adaptable to a range of environments.

(50) When using conventional techniques and mechanisms, industrial autonomous mobile robots avoid people and typically treat them like objects. In contrast, various embodiments described herein provide for autonomous mobile robots that employ semantic perception to differentiate people from static objects and move around them intelligently. An autonomous mobile robot may thus perform and/or facilitate human-centric operations such as zone picking, human following, wave picking, a virtual conveyor belt, and user training. Such operations can increase human

engagement and reduce the autonomous mobile robot's impact on foot traffic, for instance when its work is unrelated to people nearby.

(51) When using conventional techniques and mechanisms, industrial autonomous mobile robots are difficult to troubleshoot, requiring trained employees or remote support resources to resolve issues. In contrast, various embodiments described herein provide for issue resolution by individuals using the autonomous mobile robots rather than experts with specialized training.

(52) When using conventional techniques and mechanisms, industrial autonomous mobile robots typically provide limited interaction mechanisms. In contrast, various embodiments described herein provide for various types of user interaction mechanisms. For example, a user may interact with the autonomous mobile robot via a touch screen display and force-sensitive handlebars. Using such techniques, individuals may perform tasks such as moving heavy loads, teaching a fleet of autonomous mobile robots about new locations, and resolving issues without interacting with technical support services.

(53) When using conventional techniques and mechanisms, autonomous mobile robots operate using centralized and cloud computing system architectures that increase cost and latency to the robots' ability to respond to rapidly changing warehouse environments. In contrast, various embodiments described herein provide for arms that employ localized processing systems such as neural network architectures. Such approaches provide for lower latency and improved performance, increasing the safety of the autonomous mobile robot and rendering it more responsive to both people and potential hazards in a physical environment.

(54) When using conventional techniques and mechanisms, many industrial autonomous mobile robots rely on expensive LIDAR sensors that observe only a narrow slice of the surrounding environment in limited detail. In contrast, various embodiments described herein provide for autonomous mobile robots with detailed, three-dimensional views of the surrounding environment. Such configurations provide for greater safety, smarter movement and coordination, and deeper data-enabled interactions.

(55) When using conventional techniques and mechanisms, autonomous mobile robots and automated guided vehicles treat people and dynamic objects (e.g., forklifts) as static obstacles to be avoided. In contrast, various embodiments described herein provide for autonomous mobile robots that differentiate between persistent, temporary, and in-motion objects, interacting with them fluidly and efficiently.

(56) When using conventional techniques and mechanisms, an autonomous mobile robot cannot visually distinguish between different individuals. In contrast, various embodiments described herein provide for autonomous mobile robots that can respond to requests from particular individuals and navigate around an environment in more fluid, less disruptive ways. For instance, an autonomous mobile robot may be configured to follow a particular person around a warehouse environment upon request.

(57) System Architecture

(58) FIG. 1 illustrates a perspective view of an autonomous robot **100**, configured in accordance with one or more embodiments. The autonomous robot **100** includes a drive assembly **102**, a payload **108**, and a force sensing assembly **110**.

(59) In some embodiments, the drive assembly **102** may include one or more drive units **104** and one or more payload support element **106**. Each of the one or more drive units **104** may include one or more powered wheels. Each of the one or more drive units **104** may be configured to be operated, jointly or independently, to power autonomous robot **100** and provide movement to autonomous robot **100** in a backdrivable and holonomic manner.

(60) According to various embodiments, the payload support element **106** may be one or more support features (e.g., castor wheels, sliding pads, and/or other structures that may provide stability while accommodating movement). The payload support element **106** may be disposed within portions of drive assembly **102** and/or coupled to portions of the payload **108** to provide stability

for autonomous robot **100**. In various embodiments, the payload support element **106** may be disposed or coupled to any portion of the drive assembly **102** and/or the payload **108** to provide stability. As described herein, “coupled” may refer to direct or indirect (e.g., with intermediate elements) relationships between elements while “connected” may refer to direct (e.g., with no intermediate elements) relationships between elements.

(61) In some embodiments, the payload support element **106** may provide sufficient support for the payload **108** to allow for the one or more drive units **104** to be positioned in a manner to provide for predictable backdrivable and holonomic movement. For instance, the payload support element **106** may provide for stability while the payload **108** (which may include, for example, a shelf) is loaded or unloaded and/or while the autonomous robot **100** is in motion.

(62) The drive assembly **102** may be configured to couple to the payload **108** to move the payload **108**. According to various embodiments, the drive assembly **102** may couple to the payload **108** via any technique. For example, one or more openings on a body, such as one or more portions of payload **108**, may be inserted into one or more openings disposed within the body of drive assembly **102**. As another example, one or more mechanical fasteners such as bolts, screws, and/or rivets may be employed. As still another example, permanent or semi-permanent techniques such as welding or adhesives may be used. As such, drive assembly **102** may be a module that may, in some embodiments, be coupled to any number of different versions of the payload **108**.

(63) In some configurations, the payload **108** may be a commercially available (e.g., off-the-shelf) utility body, such as a shelf. Alternatively, the payload **108** may be customized for use with the drive assembly **102**.

(64) According to various embodiments, the payload **108** may include any tool or assembly that may assist in operations. For example, the payload **108** may include one or more elements of a cart (which may include a mounted shelf), a mounted robot, a container box, and/or other such item. While description may be provided in the manner of autonomous carts and shelves, other embodiments of payload **108**, such as assembly robots, are within the scope of the disclosure.

(65) In some embodiments, the force sensing assembly **110** may include, for example, a vertically oriented handle (e.g., a handle with a major axis that is within 10 degrees of vertical) coupled to the autonomous robot **100** and communicatively coupled to the drive assembly **102**. Other embodiments of the force sensing assembly **110** may include a handlebar oriented in another orientation (e.g., a horizontally oriented handle within 10 degrees of horizontal) a force sensing base (e.g., a base, such as the base of drive assembly **102**, configured to receive input from a foot of a user) of autonomous robot **100**, and/or other such mechanism or technique configured to receive directional input from a user. Such input may, for example, allow for the distinguishing of different types of inputs, such as inputs that are intended to cause the autonomous robot **100** to translate in a certain direction as well as inputs that are intended to cause the autonomous robot **100** to rotate in a certain direction.

(66) In some embodiments, the robot may include one or more sensors for supporting whole body force sensing. Using a whole body force sensing approach, force exerted anywhere on the robot can be detected, even if not exerted on a handlebar connected with a force sensor. For example, the robot's drive unit may detect a force exerted on the robot by comparing a direction and magnitude of motion of the robot compared to an instruction sent to the drive assembly to estimate a force exerted on the robot outside of the handlebar. As another example, the robot may be configured to exert a force to support backdriveability in which the robot moves in a direction of a force exerted on the robot so as to negate the force felt by the robot.

(67) In some implementations, the force sensing assembly **110** may be configured to provide operating instructions to the drive assembly **102**. That is, a user may manipulate the force sensing assembly **110** and appropriate operating instructions may be determined (e.g., by a controller disposed within the force sensing assembly **110** and/or coupled to the force sensing assembly **110** and configured to receive signals from the force sensing assembly **110**) for drive assembly **102**.

Such operating instructions may be communicated to the drive assembly **102**.

(68) In some embodiments, the force sensing assembly **110** may be a force sensing handlebar assembly that is positioned between a human operator and the payload **108** to significantly reduce the effort involved in moving the payload **108** by operating drive assembly **102** via commands determined by manipulation of the force sensing assembly **110**. The force sensing assembly **110** may, thus, operate the drive assembly **102** to push, pull, and/or rotate the autonomous robot **100** and, thus, payload **108**.

(69) In various embodiments, the force sensing assembly **110** may be positioned on various areas of the autonomous robot **100**. For instance, the force sensing assembly **110** may be positioned along the top of autonomous robot **100**, along the base of the autonomous robot **100**, or in a different location. Signals from manipulation of the force sensing assembly **110** may be communicated to the drive assembly **102** in a wired or wireless fashion.

(70) In some embodiments, vertical orientation of a force sensing handlebar may allow for ergonomic improvements for user interactions with the autonomous robot **100**. For example, a human operator may instinctively grab and manipulate items with a vertically oriented hand (e.g., with the thumb of the hand located at the top). Additionally, vertical orientation allows for intuitive rotational control of the autonomous robot **100** as the rotational controls may mimic the wrist rotation of the user.

(71) FIG. 2 illustrates a perspective view of an autonomous robot **200** with a horizontal handlebar, configured in accordance with one or more embodiments. The autonomous robot **200** that includes drive assembly **202** with one or more drive units **204**, a payload support element **206**, a payload **208**, and a force sensing assembly **210**. In many respects, the autonomous robot **200** is substantially similar to the autonomous robot **100** shown in FIG. 1. However, as shown in FIG. 2, force sensing assembly **210** may include a horizontal handlebar. The horizontal handlebar of FIG. 2 may include sensors, as described herein, disposed at one or both ends (e.g., the horizontal ends) and/or other portions of the handlebar. In certain embodiments, translational pushes on the horizontal handlebar may cause autonomous robot **200** to translate, but twisting of the horizontal handlebar (e.g., around a vertical axis such that, for example, one end of the handlebar may be moved “forward” while the other end may be moved “backward”) may cause rotation of autonomous robot **200**.

(72) FIG. 3 illustrates an additional configuration of an autonomous mobile robot **300**, configured in accordance with one or more embodiments. The autonomous mobile robot **300** includes a base unit and drive assembly **302**, a chassis **304**, a sensor unit **306**, a user interaction unit **308**, and a communication channel **310**.

(73) According to various embodiments, the base unit **302** may be configured with an omnidirectional and/or holonomic drive assembly. The drive assembly may include elements such as one or more wheels, treads, motors, controllers, batteries, and/or other components. In some configurations, the base unit and drive assembly **302** may include a force sensor such as a whole-robot force sensor. Alternatively, or additionally, such a force sensor may be included in a different portion of the autonomous mobile robot **300**, such as within the user interaction unit **308**. The base unit **302** may also include a bump sensor configured to detect an impact.

(74) According to various embodiments, the chassis **304** may include one or more rigid members providing physical support and connection between and among other components of the robots. For instance, the chassis **304** may be composed of one or more rods, shelves, bins or other elements. In some configurations, some or all of the chassis **304** may be composed of components from standardized shelving units or carts.

(75) According to various embodiments, various types and configurations of chassis may be used. In some embodiments, the chassis may be composed in part of a commodity shelving unit. For instance, in some configurations the commodity shelving unit may be 48 inches long, 38 inches wide, and 73 inches tall.

(76) According to various embodiments, the sensor unit **306** may include one or more sensors

configured to sense the physical environment in which the autonomous mobile robot **300** is situated. For instance, in particular embodiments the sensor unit **306** may include four stereo pairs of visible light cameras arranged with one on each of four sides of the robot providing 360-degree or near 360-degree visual coverage. However, depending on the configuration, various numbers and types of sensors may be employed. Examples of such sensors may include, but are not limited to: visible light cameras, infrared cameras, time-of-flight depth sensors, structured light depth sensors, RADAR sensors, LIDAR sensors, microphones, and chemical sensors. In addition to one or more sensors, the sensor unit **306** may include other elements such as one or more autonomous mobile robot controllers or computing units, one or more communication interfaces for communicating with other computing devices, and/or one or more digital display screens for displaying information.

(77) According to various embodiments, the user interaction unit **308** may include one or more elements for facilitating user interaction. For example, the user interaction unit **308** may include a display (e.g., a touch-screen display) for presenting information and/or receiving user input. As another example, the user interaction unit **308** may include a force-sensitive handlebar configured to force exerted on the handlebar. The force detected may include degree, direction, and/or rotational elements. As still another example, the user interaction unit **308** may include a barcode scanner or other sensor.

(78) According to various embodiments, the communication channel **310** may include one or more cables and/or busses for transmitting power, sensor data, instructions, and/or other electronic signals between different components of the robot. For instance, the communication channel **310** may include routing for accessor cables, lighting, put-to-light taps, pick-from-light taps, and/or other components.

(79) It should be noted that FIG. 3 illustrates only one example of a configuration of an autonomous mobile robot. However, an autonomous mobile robot may be configured in a different manner in accordance with one or more embodiments. For example, an autonomous mobile robot may include more than one handlebar. As another example, a handlebar may be configured in a different orientation, such as a vertical orientation. As another example, a handlebar may be integrated into the chassis. As still another example, one or more elements of the sensor unit may be distributed elsewhere on the chassis **304** or base unit **302**. As yet another example, the chassis may be arranged with different configurations of shelving or other components.

(80) FIG. 4 illustrates a disassembled view of an autonomous mobile robot, configured in accordance with one or more embodiments. As shown in FIG. 4, the autonomous mobile robot may be separated into components for easy assembly, including a charging dock **402**, a base unit and drive assembly **404**, a payload **406**, a head unit **408**, a light bar **410**, and a communication channel **412**. For example, the components may be shipped in a disassembled state and assembled on site. In some embodiments, a mobile robot may dock with the charging dock **402** to charge a battery included within the base assembly **404**.

(81) FIG. 5 illustrates a communication architecture diagram **500** of the autonomous mobile robot, configured in accordance with one or more embodiments. The communication architecture diagram **500** is conceptually divided into regions. A sensor unit region **502** corresponds with the sensor unit **306**. A user interaction region **504** corresponds with the user interaction unit **308**. A base unit region **506** corresponds with the base unit and drive assembly **302**. A drive unit region **508** corresponds with one or more drive units within the base unit and drive assembly **102**.

(82) According to various embodiments, the sensor unit region **502** may include a main processing unit **510**, a communication interface **512**, one or more sensors **514**, and/or one or more marquees **516**. The main processing unit **510** may be a computing device such as an AGX Orin provided by Nvidia. The communication interface **512** may include a hardware radio or other device facilitating communication using a communication protocol such as WiFi, Bluetooth, and/or cellular. The sensors **514** may transmit sensor data to the main processing unit.

(83) In some embodiments, the main processing unit **510** may be configured to process and/or instruct the sensors **514**. For instance, the main processing unit **510** may be configured to determine a model of a physical environment based on camera data from one or more visible light cameras included in the sensors **514**.

(84) According to various embodiments, the base unit region **506** includes a main board **524**, which includes one or more processors and/or other components. The main board **524** facilitates communication between and among other components. The base unit region **506** also includes one or more force sensors **526** and a power distribution board **528**. The power distribution board **528** communicates with one or more battery systems **530**, a power dock interface **532**, an on button **534**, and an electronic stop interface **536**.

(85) According to various embodiments, the user interaction region **504** may include one or more end points for interacting with user interface devices such as one or more touch sensors **518**, lighting elements **516**, touch screens **520**, barcode scanners **522**, and/or other such devices. Such devices may communicate with the main processing unit **510** and/or the main board **524**.

(86) According to various embodiments, the drive unit region **510** may communicate with motor driver boards **538** and **540** corresponding to different drive units within the autonomous mobile robot, which may have one, two, or more drive units. Each drive unit may correspond to one or more wheels, treads, or other mechanisms for locomotion. The motor driver boards may communicate with the encoders **542** and **544** and one or more motors **546** and **548**. For instance, the encoders **542** and **544** may be absolute encoders, and the motors **546** and **548** may be brushless DC motors.

(87) The communication architecture diagram **500** is one example of a possible configuration of components within the autonomous mobile robot **100**, provided for the purpose of illustration. According to various embodiments, various arrangements and combinations of components may be employed in a manner consistent with techniques and mechanisms described herein.

(88) Additional details regarding a mechanical drive unit that may be used in conjunction with techniques and mechanisms described herein are described in U.S. patent application Ser. No. 18/622,640 by Davey et al, filed Mar. 29, 2024, titled “Autonomous Robot Double Drive Assembly”, and in U.S. Provisional Patent Application No. 63/561,023 by Davey et al, filed Mar. 5, 2024, titled “Autonomous Robot Double Drive Assembly”, both of which are incorporated herein by reference in their entirety and for all purposes.

(89) FIG. **6** is a block diagram of a computing device, configured in accordance with one or more embodiments. According to various embodiments, a system **600** suitable for implementing embodiments described herein includes a processor **601**, a memory module **603**, a storage device **605**, an interface **611**, and a bus **615** (e.g., a PCI bus or other interconnection fabric.) System **600** may operate as variety of devices such an autonomous mobile robot, a remote server configured as a fleet manager, or any other device or service described herein. Although a particular configuration is described, a variety of alternative configurations are possible. The processor **601** may perform operations such as those described herein with respect to the various devices and methods. Instructions for performing such operations may be embodied in the memory **603**, on one or more non-transitory computer readable media, or on some other storage device. Various specially configured devices can also be used in place of or in addition to the processor **601**. The interface **611** may be configured to send and receive data packets over a network. A computer system or computing device may include or communicate with a monitor, printer, or other suitable display for providing any of the results mentioned herein to a user.

(90) Any of the disclosed implementations may be embodied in various types of hardware, software, firmware, computer readable media, and combinations thereof. For example, some techniques disclosed herein may be implemented, at least in part, by non-transitory computer-readable media that include program instructions, state information, etc., for configuring a computing system to perform various services and operations described herein. Examples of

program instructions include both machine code, such as produced by a compiler, and higher-level code that may be executed via an interpreter. Instructions may be embodied in any suitable language such as, for example, Java, Python, C++, C, HTML, any other markup language, JavaScript, ActiveX, VBScript, or Perl. Examples of non-transitory computer-readable media include, but are not limited to: magnetic media such as hard disks and magnetic tape; optical media such as flash memory, compact disk (CD) or digital versatile disk (DVD); magneto-optical media; and other hardware devices such as read-only memory (“ROM”) devices and random-access memory (“RAM”) devices. A non-transitory computer-readable medium may be any combination of such storage devices.

(91) Drive Assemblies

(92) Various configurations of autonomous mobile robots may be configured with various types of drive assemblies. FIG. 7 illustrates a diagram of a drive unit **700**, configured in accordance with one or more embodiments. FIG. 8 illustrates a different view of the autonomous mobile robot that includes two drive units **700**.

(93) The drive unit **700** includes a turntable unit **702** that turns about an axis **704**. The drive unit **700** also includes one or more wheels **706** and one or more motors. The wheels **706** are offset from the axis **704** and independently controllable via the motor. The motor may power one or more of the wheels using power supplied by a battery, which may be located inside of the drive unit or outside of the drive unit and may be coupled directly or indirectly with the motor, for instance via a slip ring. The drive unit **700** may include one or more unpowered supports, such as the unpowered freely mobile caster wheel **710**, for instance to provide additional stability.

(94) In the configuration shown in FIG. 8, each drive unit rotates freely around an axis **704**. That is, the rotation around the axis **704** is unpowered. The autonomous mobile robot can be caused to move in any direction or rotated by applying power from the motors to the wheels **706**.

(95) In FIG. 8, an autonomous mobile robot is further supported by one or more support elements **802**. For instance, a support element may be a freely spinning and unpowered caster wheel, a slider, or some other component. The support elements may provide additional stability to the autonomous mobile robot.

(96) In some embodiments, an autonomous mobile robot may be equipped with multiple support elements **802**. For instance, an autonomous mobile robot may be equipped with four freely spinning and unpowered caster wheels located at approximately the four corners of the autonomous mobile robot.

(97) According to various embodiments, an autonomous mobile robot may be equipped with one or more drive units. For example, FIG. 9A illustrates a configuration that includes a single drive unit, FIG. 9B illustrates a configuration that includes a double drive unit, and FIG. 9C illustrates a configuration that includes a triple drive unit. In some implementations, including more than one drive unit may include performance on one or more dimensions. For instance, moving from a single drive unit to a double drive unit may increase backdrivability, decrease torque requirements, decrease the number of unique parts, and decrease the diameter needed for each of the drive units.

(98) In some embodiments, an autonomous mobile robot may be equipped with one or more drive units of a different type from that shown, such as one or more drive units that employ one or more Mecanum wheels.

(99) FIG. 10A, FIG. 10B, and FIG. 10C illustrate perspective views of portions of an autonomous mobile robot. As shown in FIG. 10A, FIG. 10B, and FIG. 10C, a base unit **102** that includes a drive assembly may be removable from the autonomous mobile robot.

(100) Force Sensing

(101) FIG. 11A and FIG. 11B illustrate handlebars for guiding the autonomous mobile robot **100**, configured in accordance with one or more embodiments. Depending on the configuration, handlebars may be arranged in any of various ways.

(102) In some embodiments, as shown in FIG. 11A, a handlebar **1102** may be arranged in a

substantially vertical orientation. Alternatively, or additionally, a handlebar may be arranged in a substantially horizontal orientation, as shown in FIG. **11B**. As still another possibility, a handlebar may be arranged in a different orientation

(103) In some embodiments, a handlebar **1102** may be integrated with the chassis, as shown in FIG. **11A**. Alternatively, or additionally, a handlebar **1104** may be set off from the chassis, as shown in FIG. **11B**.

(104) In some embodiments, an autonomous mobile robot may be equipped with a single handlebar. Alternatively, in some configurations an autonomous mobile robot may be equipped with more than one handlebar, as shown in FIG. **11A** and FIG. **11B**.

(105) In some embodiments, the handlebars may be used to detect and amplify force applied to the robot. For example, torque as the handlebar is twisted may be detected and used to instruct the drive unit to rotate the robot around the axis of the handlebar. As another example, translational force as the handlebar is pushed may be detected and used to instruct the drive unit to move the robot in the direction of the force, effectively amplifying the force.

(106) FIG. **12** illustrates a force sensor **1200**, configured in accordance with one or more embodiments. The force sensor **1200** may be used to detect force exerted on one or more of handlebars, for instance a handlebar configured as shown in FIG. **11**. The force sensor **1200** includes a hall effect sensor **1202**, a spring gasket **1204**, and one or more magnets **1206**.

(107) In some embodiments, a bar passes through the spring gasket **1204**. When force is exerted on the bar and then removed, the spring gasket **1204** causes the bar to return to its original central position.

(108) In some embodiments, the one or more magnets **1206** may be arranged to as to generate a magnetic field detected by the hall effect sensor **1202**, which may detect disruptions to the magnetic field corresponding with force exerted in one, two, three, or four dimensions. For instance, the hall effect sensor **1202** may detect disruptions to the magnetic field corresponding with force exerted in the x-axis, the y-axis, the z-axis, and/or a rotational force.

(109) In some embodiments, the hall effect sensor **1202** may translate the detected disruptions into force sensor data. The force sensor data may identify a direction of the force in one, two, or three translational dimensions and/or a fourth rotational dimensions. Alternatively, or additionally, the force sensor data may identify a magnitude corresponding with a translational and/or rotational force.

(110) According to various embodiments, an autonomous mobile robot may be equipped with any of various kinds of force sensors. Examples of force sensors may include, but are not limited to: Hall effect sensors, optical sensors, capacitive touch sensors, button switch sensors, break beam sensors, force sensitive resistors, and force sensitive switches.

(111) According to various embodiments, an autonomous mobile robot may be equipped with various numbers and types of force sensors. For example, an autonomous mobile robot **100** may be equipped with force sensors located at some or all of a set of vertical poles included in the chassis and/or at some or all of handlebars coupled with the autonomous mobile robot.

(112) According to various embodiments, a force sensor may be located in any of various locations on an autonomous mobile robot. For example, a force sensor may be located at the handlebar itself. Alternatively, or additionally, a force sensor may be located at the robot chassis, for instance along a vertical pole included in the chassis.

(113) According to various embodiments, force sensor location may give rise to one or more tradeoffs. For instance, locating a force sensor at the chassis may provide for increased flexibility, since the robot may be grabbed and maneuvered from different sides. However, locating a force sensor at the chassis may cause the user to feel the payload mass when exerting force on the robot, and may increase hysteresis. In contrast, locating a force sensor at the handlebar may hide the payload mass from the user. However, locating a force sensor at the handlebar may limit the user to grabbing the cart from the handlebar.

(114) FIG. 13 illustrates a perspective view of a force sensing handle assembly **1310**, configured in accordance with one or more embodiments. The handle assembly **1310** includes a handlebar **1312** and a housing **1328**. The handlebar **1312** may be a vertically or horizontally oriented handlebar configured to allow a user to grasp the handlebar to provide operating instructions to the autonomous robot. In various embodiments, the housing **1328** may be configured to interface (e.g., mount) to one or more other portions of the autonomous robot, such as to the payload **108**. In certain such embodiments, the housing **1328** may be configured to couple to various different portions and/or versions of the autonomous robot.

(115) The ends of the handlebar **1312** may be disposed within the housing **1328**. The housing **1328** may include various sensors as well as fixtures that the handlebar **1312** is coupled to. The internal elements of the housing **1328** and the handlebar **1312** are further illustrated in FIG. 14. FIG. 14 illustrates a perspective view of portions of a force sensing handlebar, configured in accordance with one or more embodiments. FIG. 14 illustrates a handle assembly **1410**, which may not include the housing **1328**.

(116) In FIG. 14, the handle assembly **1410** includes a handlebar **1412**. The handlebar **1412** includes a first end **1432A** and a second end **1432B** on opposite distal portions. The ends of the handlebar **1412** may be coupled to various fixtures. For example, the first end **1432A** of the handlebar **1412** may be coupled to the fixture **1414A** while the second end **1432B** of the handlebar **1412** may be coupled to the fixture **1414B**.

(117) In certain embodiments, the handlebar **1412** may be coupled to the fixtures **1414A** and/or **1414B** via compliant material **1430A** and/or **1430B**. The compliant material **1430B** is not entirely visible in FIG. 14, but an ordinal indicator for **1430B** indicates where the compliant material **1430B** is located. The compliant material **1430B** is coupled to the fixture **1414B** in a similar manner to the manner that the compliant material **1430A** is coupled to the fixture **1414A**. The compliant material **1430A** and/or **1430B** may be a form or material, such as a spring or bushing made from an elastomer, rubber, metal, and/or other material, that allows the position of handlebar **1412** to change relative to the fixtures **1414A** and/or **1414B** in response to force applied to the handlebar **1412** by a human operator. In various embodiments, the compliant materials **1430A** and/or **1430B** may be coupled via casting, friction fit, fasteners, adhesives, and/or another such technique that may allow for the joining of two items (e.g., two items of different materials).

(118) In some embodiments, the fixtures **1414A** and/or **1414B** may be coupled to another portion of the autonomous robot. For instance, the fixtures **1414A** and/or **1414B** may be coupled to the autonomous robot via another portion of the handle assembly **1310** that may then be coupled to the autonomous robot. The fixtures **1414A** and **1414B** may thus be coupled to the housing **1328** to hold the handlebar **1412** in a position relative to the housing **1328**.

(119) In some embodiments, the fixtures **1414A** and/or **1414B** may be directly connected to the autonomous robot (e.g., via any type of direct connection such as adhesives, fasteners, welding, and/or via fasteners or other removable techniques). The compliant material **1430A** and/or **1430B** may, thus, allow for the handlebar **1412** to translate and/or rotate relative to the fixtures **1414A** and/or **1414B** in response to force applied by the user. Furthermore, the fixtures **1414A** and/or **1414B** in combination with the compliant material **1430A** and/or **1430B** may be configured to hold the handlebar **1412** in a fixed position (e.g., neutral position) when no force is applied to handlebar **1412**.

(120) The handle assembly **1410** includes a first sensor **1446A** and a second sensor **1446B**. The first sensor **1446A** includes a first sensor first portion **1416A** and a first sensor second portion **1418A**. The second sensor **1446B** includes a second sensor first portion **1416B** and a second sensor second portion **1418B**. The first sensor first portion **1416A** and the second sensor first portion **1416B** may be coupled to handlebar **1412**.

(121) In some embodiments, the first sensor first portion **1416A** and the second sensor first portion **1416B** proximate to the opposite distal ends (e.g., the first end **1432A** and the second end **1432B**)

of the handlebar **1412**. Alternatively, the first sensor first portion **1416A** and the second sensor first portion **1416B** may be coupled to another portion of handlebar **1412**.

(122) In some implementations, the first sensor second portion **1418A** and the second sensor second portion **1418B** may be coupled to portions of handle assembly **1410** and/or autonomous robot that handlebar **1412** may be configured to move relative to. That is, the first sensor second portion **1418A** and the second sensor second portion **1418B** may be coupled to, for example, the housing **1328**, the fixtures **1414A** and **1414B**, respectively, and/or another portion of autonomous robot. Thus, the first sensor second portion **1418A** and the second sensor second portion **1418B** may be held in a “fixed” position (e.g., fixed relative to another portion of autonomous robot **100** such as payload **108**) so that movement of the handlebar **1412** may cause the first sensor first portion **1416A** and the second sensor first portion **1416B** to move relative to the first sensor second portion **1418A** and the second sensor second portion **1418B**.

(123) In some embodiments, relative movement of the first sensor first portion **1416A** to the first sensor second portion **1418A** and the second sensor first portion **1416B** to the second sensor second portion **1418B** may allow for a determination as to whether a human operator is pushing on or rotating the handlebar **1412**. Based on the human operator's interaction with the handlebar **1412** (e.g., whether the user is pushing on or rotating handlebar **1412**), the autonomous robot may be driven in different manners. Operation of the autonomous robot is discussed in additional detail throughout the application, for instance with respect to FIG. **19** through FIG. **26**.

(124) In some embodiments, the first sensor first portion **1416A** and the first sensor second portion **1418A** and the second sensor first portion **1416B** and the second sensor second portion **1418B** may be offset in different positions (e.g., different positions along a vertical axis for vertically oriented handlebars or different positions along a horizontal axis for horizontally oriented handlebars) to allow for distinguishing of translational and rotational operating instructions.

(125) In some embodiments, the first sensor first portion **1416A** may be configured to interact with the first sensor second portion **1418A**, while the second sensor first portion **1416B** may be configured to interact with the second sensor second portion **1418B**. For example, first sensor second portion **1418A** first sensor second portion **1418A** may be configured to sense movement of first sensor second portion **1418A** first sensor first portion **1416A** relative to first sensor second portion **1418A** first sensor second portion **1418A**. The second sensor second portion **1418B** may be configured to sense movement of the second sensor first portion **1416B** relative to the second sensor second portion **1418B**. Such relative movement may be, for example, due to deflection of the compliant materials **1430A** and/or **1430B** from forces applied to the handlebar **1412**.

(126) In some implementations, the first sensor first portion **1416A** and the second sensor first portion **1416B** include magnets while the first sensor second portion **1418A** and second sensor second portion **1418B** may include hall effect sensor modules (e.g., 1-axis, 2-axis, or 3-axis hall effect sensors) that includes one or a plurality of hall effect sensors. The hall effect sensor modules may be configured to sense the respective movement of the magnets, or vice versa. Utilizing a plurality of paired sensor portions that are offset from each other allows for measurement of twist theta in addition to that of relative displacement, allowing for estimates of applied force (e.g., linear force) as well as applied torque (e.g., twisting force). Other embodiments may include other types of sensors for the first sensor first portion **1416A**, the second sensor first portion **1416B**, the first sensor second portion **1418A**, and/or the second sensor second portion **1418B**, such as optical flow sensor, optical or magnetic encoders, potentiometers, ultrasonic sensors, and/or other such sensors.

(127) In some embodiments, the handlebar assembly may include a break beam sensor. The break beam sensor may transmit a signal when a beam of light is broken and/or reestablished. Such a sensor may be used to detect when a handlebar is grasped and/or released.

(128) FIG. **15** illustrates a top view of a force sensing handlebar for an autonomous robot in a first manipulated position, configured in accordance with one or more embodiments. FIG. **15** illustrates a linear force **1522** applied to the handlebar **1512** of the handle assembly **1510**. In certain

embodiments, the linear force **1522** applied to handlebar **1512** may result in drive assembly **102** linearly moving (e.g., translating) the autonomous robot, according to the techniques and mechanisms described herein. The speed of the linear movement may be dependent on the magnitude of the detected force applied to the handlebar **1512**.

(129) Movement of the first sensor first portion **1516A** may be determined relative to the sensor axis **1520A**. When no force is applied to the handlebar of the handle assembly **1510**, the first sensor first portion **1516A** may be determined to be disposed at the center, or proximate the center (e.g., within a set degree of tolerance, such as within a few millimeters), of the sensor axis **1520A** and may be determined to be disposed in a neutral position. The first sensor second portion **1518A** may be calibrated to determine the position of the first sensor first portion **1516A**. The first sensor second portion **1518A** may be calibrated such that when the first sensor first portion **1516A** is disposed in the neutral position (e.g., at the center or proximate the center of sensor axis **1520A**), the first sensor second portion **1518A** may determine that there is no relative movement of the first sensor first portion **1516A**.

(130) Additionally, the first sensor second portion **1518A** may be configured to detect movement of the first sensor first portion **1516A** along two axes (e.g., XA and YA). Thus, when force is applied to the handlebar, in certain instances, the first sensor first portion **1516A** may move along the XA and/or YA axes (e.g., in the positive or negative XA and/or YA directions relative to the sensor axis **1520A**) and such movement may be detected by the first sensor second portion **1518A**. Similarly, movement of the second sensor first portion **1516B** may be determined by the second sensor second portion **1518B**, relative to the sensor axis **1520B**, the center of which may be a neutral position that the second sensor second portion **1518B** is calibrated towards.

(131) The linear force **1522** may be applied to the handlebar **1512**. Due to the application of the linear force **1522**, the first sensor first portion **1516A** may move relative to the first sensor second portion **1518A**. Such movement may be determined as positive or negative according to defined axes. Thus, in certain embodiments, movement in the positive XA direction and the positive YA direction may be classified as positive magnitude, while movement in the opposite direction may be classified as negative magnitude. Similarly, the second sensor second portion **1518B** may be configured to determine movement of the second sensor first portion **1516B** in the positive and negative XB and YB directions, as shown. In certain embodiments, the positive directions of XA and XB may be in the same direction while the positive directions of YA and YB may be in opposite directions. The positive and negative directions may allow for the determination of whether the handlebar **1512** is translating or rotating. Other orientations of the axes may be possible in other embodiments.

(132) As shown in FIG. **15**, the linear force **1522** applied to the handlebar **1512** may cause the reaction **1524A** for the first sensor first portion **1516A** (e.g., the reaction **1524A** may include movement of the first sensor first portion **1516A** in the positive XA direction) and may cause the reaction **1524B** for the second sensor first portion **1516B** (e.g., the reaction **1524B** may be movement of the second sensor first portion **1516B** in the positive XB direction). The reactions of the first sensor first portion **1516A** and the second sensor first portion **1516B** may be detected by the first sensor second portion **1518A** and the second sensor second portion **1518B**, respectively. Based on the determination that the first sensor first portion **1516A** is moving in the positive XA direction and the second sensor first portion **1516B** is moving in the positive XB direction, a determination may be made (e.g., by a controller as described herein) that the force **1522** is causing the handlebar **1512** to translate as both first sensor first portion **1516A** and the second sensor first portion **1516B** may both be determined to be moving in the same direction, along the same vector, and/or with the same magnitude of movement.

(133) FIG. **16** illustrates a top view of a force sensing handlebar for an autonomous robot in a second manipulated position, configured in accordance with one or more embodiments. FIG. **16** illustrates torque **1626** applied to the handlebar **1612** of the handle assembly **1610**. The torque **1626**

may be, for example, a twisting motion applied by a user. Application of the torque **1626** may cause the orientation of the handlebar **1612** to accordingly twist, resulting in the reaction **1624A** for the first sensor first portion **1616A**, which may be movement of the first sensor first portion **1616A** in at least the positive XA direction, and the reaction **1624B** for the second sensor first portion **1616B**, which may be movement of the second sensor first portion **1616B** in at least the negative XB direction. In certain embodiments, the first sensor first portion **1616A** may additionally or alternatively move in the negative YA direction and the second sensor first portion **1616B** may additionally or alternatively move in the negative YB direction.

(134) Based on the determination that the first sensor first portion **1616A** is moving in the positive XA direction and the second sensor first portion **1616B** is moving in the negative XB direction, a determination may be made that the torque **1626** is causing the handlebar **1612** to rotate. The drive assembly **102** may be then operated to cause autonomous robot **100** to rotate orientations. Other embodiments may determine other types of rotation (e.g., with the first sensor first portion **1616A** moving in the negative XA direction and the second sensor first portion **1616B** moving in the positive XB direction).

(135) In the examples of FIGS. **15** and **16**, in addition to determination of the type and direction of movement of the handlebar, the magnitude of the force and/or torque applied may also be determined. Such magnitude may be determined based on the stiffness factor of the fixtures **1414A** and **1414B** and/or the compliant materials **1430A** and **1430B**. Thus, it is appreciated that disposing the first sensor portions as close as possible to their respective fixtures may provide a simpler technique for determination of movement of the handlebar at the position of the respective fixtures.

(136) In some embodiments, the magnitude of the force and/or torque may be determined based on the relationship of $F=kx$ and $x=Bk/m$, where k is the stiffness factor from the fixture and/or compliant materials and m is a factor relating distance to magnetic flux output from a hall effect sensor (e.g., as the magnet of the first sensor portion is disposed farther away from the hall effect sensor of the second sensor portion, the flux output by the hall effect sensor changes, allowing for determination of the distance between the first sensor portion and its respective second sensor portion). The lumped parameter k/m may be an empirically determined factor relating force and magnetic flux. In certain embodiments, such as embodiments where the hall effect sensor is one or more 3-axis (3 degree of freedom) hall effect sensors, the differences in magnetic flux based on the orientation of the magnet may be detected and, accordingly, whether the magnet is moving in the X or Y direction may be determined.

(137) FIG. **17** illustrates a perspective view of an autonomous robot with a force sensing base, configured in accordance with one or more embodiments. FIG. **17** illustrates an autonomous robot **1700** that includes a drive assembly **1702** with a drive unit **1704**, a payload support element **1706**, a payload **1708**, and the force sensing assembly **1710**. A human operator may interact with the force sensing base by, for example, pushing or pulling on the force sensing base with a variety of different force input directions to cause the autonomous robot **1700** to translate and/or rotate (e.g., based on pushes that are orthogonal to the force sensing assembly **1710** to cause translational movement or pushes that are on the bias, such as within ± 15 degrees of 45 degrees, to the force sensing assembly **1710** to cause rotational movement). Furthermore, as the sensors are agnostic as to whether the payload **1708** or the force sensing assembly **1710** is being pushed/pulled due to the sensors being configured to determine relative movement between the payload **1708** and the force sensing assembly **1710**, a human operator may also push or pull on the payload **1708** to translate and/or rotate the autonomous robot **1700** in a similar manner to that of interacting with the base.

(138) FIG. **18** illustrates a perspective view of portions of a force sensing base, configured in accordance with one or more embodiments. FIG. **18** illustrates an autonomous robot **1800** that includes force sensing assembly **1810** that may be a force sensing base. As shown, force sensing base **1810** may include a plurality of sensors including a first sensor **1846A** and a second sensor **1846B**, as well as additional sensors.

(139) In certain embodiments, the first sensor **1846A** may include a first sensor first portion **1816A** coupled to a portion of a payload **1808** and a first sensor second portion **1818A** coupled to a portion of a force sensing assembly **1810**. Additionally, the fixture **1814A**, which may include compliant material **1830A** configured in the same manner as that described herein, may allow for movement of the payload **1808** relative to the force sensing base **1810** (e.g., in response to user inputs). Such movement may result in the first sensor first portion **1816A** moving relative to the first sensor second portion **1818A**, due to the compliance of the compliant material.

(140) Similarly, the second sensor **1846B** may include the second sensor first portion **1816B**, coupled to the payload **1808**, and the second sensor second portion **1818B**, coupled to the force sensing assembly **1810**. Similarity or differences in the detected movement of the first sensor first portion **1816A** and the second sensor first portion **1816B**, as described herein, may result in a determination of whether the autonomous robot **1800** is moved in a translational, a rotational manner, or both.

(141) Autonomous Mobile Robot Movement

(142) FIG. **19** illustrates a method **1900** of controlling an autonomous mobile robot, configured in accordance with one or more embodiments. The method **1900** may be performed at a robot configured as described herein.

(143) According to various embodiments, the method **1900** may be used to transition between various operating modes. Examples of various operating modes are described herein. However, robots may be configured with various types of operating modes depending on the operating context.

(144) In some embodiments, a robot may operate in an autonomous mode in which the robot operates autonomously to perform a task based on instructions received from a remote computing device, such as a fleet controller, or based on internal programming logic. For example, a robot operating in autonomous mode may perform operations such as moving from one location to another location while avoiding obstacles, taking one or more actions to avoid damage or injury to objects or humans, and/or operating one or more lights, mechanical arms, conveyer belts, and/or other components.

(145) In some embodiments, a robot may operate in a manual mode in which the robot acts in a manner responsive to user input. For instance, a robot may remain stationary until force is detected at a force sensor, at which time the robot may move in a direction determined based on the detected force.

(146) In some embodiments, an operating mode may include both manual and autonomous elements. For example, while operating in manual mode, a robot may nevertheless apply a force so as to avoid colliding with a person or object, effectively overriding the user input. For instance, the robot may apply a force that repels the robot from objects and that increases in magnitude with proximity to an object. As yet another example, while operating in autonomous mode, a robot may nevertheless adjust its movement based on user input, such as force received via a force sensor. As yet another example, a robot may operate in a hybrid mode in which it guides a user along a path but nevertheless allows itself to be redirected based on user input. As still another example, a robot may operate in a following mode in which it autonomously follows a human, for instance to aid in the human's efforts to perform a task.

(147) In some embodiments, a robot may shift between operating modes based on user input. For example, a robot that detects the presence of a human may stop what it is doing and face the human. Then, if the human steps away from the robot, the robot may resume autonomous activity. As another example, a robot may operate autonomously until a human grasps the robot's handlebar, at which point the robot may enter into a manual operating model. Then, if the human releases the handlebar, the robot may resume autonomous operation, for instance after the passage of a designated period of time.

(148) At **1902**, a request to control an autonomous mobile robot is received. In some embodiments,

the request may be generated automatically at a main processing unit or other controller of the robot, for instance during an initialization process.

(149) An operating mode for the robot is determined at **1904**. According to various embodiments, an autonomous mobile robot may be configured for operation in various modes, such as forced sensing, autonomous movement, unforced sensing, and/or other modes. In some embodiments, the mode may be determined based on user input. For example, a user may touch or approach the robot to remove it from autonomous movement mode. As another example, a user may activate a button or touch screen to place the robot into forced sensing mode. In some embodiments, the mode may be determined based on instructions received via a communication interface. For instance, the robot may receive an instruction from a fleet controller to enter or leave autonomous mode. As yet another example, the robot may detect when the operator has let go, for instance via one or more capacitive, tactile, and/or force sensing sensors.

(150) Input for moving the robot is determined at **1906**. In some embodiments, the input may be received via a user interface at the robot, such as a force sensing handlebar. Alternatively, the input may be received via a communication interface, for instance from a fleet controller.

(151) Instructions for a drive mechanism for the robot are determined at **1908**. The instructions are transmitted to the drive mechanism at **1910**. The instructions may cause the robot to move in a particular direction. As discussed herein, various types of instructions are possible based on various types of input.

(152) In some embodiments, an autonomous mobile robot in force sensing mode may move in a direction of force exerted by an operator on a force sensing handlebar. For example, the robot may detect a translational direction, rotational direction, and/or magnitude of force, and then direct the drive unit to move the robot in the direction. The robot may be configured such that the operator need only apply a small amount of force despite the robot carrying a heavy load, with the robot effectively magnifying that force to move in the requested direction. According to various embodiments, various types of modifications and constraints may be applied to such a force sensing motion configuration.

(153) In some embodiments, force may be applied asymmetrically, for instance braking much more easily than accelerating. Further, the robot may sense its surroundings and adapt its instructions for safety and/or damage avoidance. For example, the robot may slow to a stop before striking a person or object, moving down a ramp, or entering a hole. As another example, the robot may slow to a stop when it detects that the operator is no longer touching the robot.

(154) A determination is made at **1912** as to whether to continue controlling the robot. According to various embodiments, the robot is controlled until it is deactivated.

(155) FIG. **20** illustrates a method **2000** for executing a robot control loop, performed in accordance with one or more embodiments. The method **2000** may be used to determine instructions for providing to a robot drive unit for the purpose of causing the robot to move through space. The method **2000** illustrates a more detailed view of operations discussed with respect to the method **1900** shown in FIG. **19**.

(156) The method **2000** is described with reference to FIG. **21**, which illustrates an architecture diagram for a control portion of a mobile robot configured in accordance with one or more embodiments. The architecture diagram includes drive control input sources **2102** through **2104**, which may provide input to a robot drive controller **2106**. The robot drive controller **2106** may determine output instructions for a robot drive unit **2110** and may provide the output instructions to the robot drive unit **2110** via a robot drive unit abstraction layer **2108**. The method **2000** may be performed at the robot drive controller **2106**, which in some embodiments may be implemented at the main board **524** shown in FIG. **5**.

(157) According to various embodiments, the method **2000** may be used with any of various types of robots and/or drive systems. Some examples of such robots and drive systems are discussed with respect to FIGS. **1-18**. However, the method **2000** may be used in conjunction with various types of

robots and/or drive systems. For example, suitable robots may or may not be autonomous, omnidirectional, and/or backdrivable.

(158) A request to control a drive unit of a robot is received at **2002**. In some embodiments, the request may be generated when the robot is activated and enters a mode in which it is controllable. The robot may be controlled based on one or more of user input, autonomous decision-making, and/or remote instructions received via a network interface from a remote system such as a fleet controller.

(159) Movement control input information is determined at **2004**. According to various embodiments, the movement control input information may include various types of input received from any of various sources, such as the sources **2102** through **2104** shown in FIG. **21**. For example, the movement control information may include user input received from a user input device such as a force sensor, joystick, or other control device. For instance, the force sensor may be attached to a force sensing handlebar as discussed throughout the application. As another example, the movement control information may include one or more configuration parameters such as a force multiplier, a virtual friction coefficient, an indication of an operating mode, or the like. As yet another example, the movement control information may include one or more functional control parameters. For instance, the robot may be operating in a mode such that it behaves as if it is moving between virtual rails.

(160) According to various embodiments, the drive unit of the robot may be controlled at least in part by transmitting an instruction determined based on user input received at a force sensor at the robot. For instance, when force is detected at a force sensing handlebar, the robot may be moved in the direction of the force. In general, a relatively larger force detected at the force sensor may correspond with a relatively higher velocity or force applied to the robot drive unit. The term “force multiplier” as used herein refers to any alteration applied to the user input to strengthen or weaken the relationship between the input force received at the force sensor and the force instruction sent to the drive unit. For a given input force, for example, a relatively larger force multiplier would yield a relatively larger increase in velocity. As discussed throughout the application, the force multiplier may be a fixed or configurable scalar, vector, or force output function that receives as inputs one or more parameters including data from the force sensor.

(161) One or more operating conditions for the robot are determined at **2006**. In some embodiments, the one or more operating conditions may include any conditions that may affect the robot's handling and control. Examples of such operating conditions may include, but are not limited to: the identity of an operator using the robot, a location of the robot, a direction in which the robot is traveling, an amount of traffic in the vicinity of the robot, a condition associated with the physical environment in which the robot is situated, and the like. For example, the detection of a potentially unsafe condition such as a wet surface may cause all robots to be placed in a safety mode in which a lower force multiplier is applied.

(162) In some embodiments, information about operating conditions may be determined by the robot itself. For instance, the robot may detect the presence of a wet surface based on a loss of traction in the drive unit. Alternatively, or additionally, such information may be received via a communication interface, for instance from a fleet controller. As still another possibility, a user may provide input indicating one or more operating conditions.

(163) A physical input force vector is determined at **2008**. In some embodiments, the physical input force vector may be determined based on user input. The force sensor force vector may identify values for force exerted in one or more dimensions at one or more force sensors. For instance, a user may provide input by exerting force on a handlebar connected with the robot. As one example, the force sensor force vector may identify values for translational and rotational forces applied to a force sensor attached to a handlebar at the robot.

(164) In some embodiments, the force sensor force vector may identify values for force exerted in one or more dimensions at one or more force sensors. For instance, the force sensor force vector

may identify values for translational (e.g., an x-dimension and a y-dimension) and rotational forces applied to a force sensor attached to a handlebar at the robot.

(165) In some embodiments, forces may be quantified in a coordinate system. In some configurations, the coordinate system may be parameterized relative to the robot. For instance, the x-direction may be treated as “forward” while the y-direction is treated as “sideways”.

Alternatively, the coordinate system may be parameterized relative to the physical environment. For instance, the x-direction may be treated as “north” or as a particular direction within a building. Additional details regarding an example of such a coordinate situation are discussed herein with respect to the force sensing handlebar assembly.

(166) In particular embodiments, determining the physical input force vector may involve imposing one or more types of smoothing. For example, the system may impose a “dead band” around zero force. In such a configuration, the robot may ignore small amounts of force applied to the force sensor. In this way, the robot may be prevented from drifting when very little force is being applied to the force sensor. Thus, small amounts of force may be mechanically treated as zero force. As another example, the system may smooth force over time, for instance to avoid jitter.

(167) In some embodiments, one or more smoothing operations may be applied in a dimension-specific manner. For instance, if a non-trivial force is being applied in the forward direction but only a trivial amount of force is being applied in the rotational direction, then the non-trivial forward movement force may be accounted for in the physical input force vector while the trivial rotational force is ignored. In this way, a user may be able to easily move the robot in one dimension without the robot drifting in another dimension. Alternatively, one or more smoothing operations may be applied in a dimensionless manner. For instance, if the sum of the magnitudes of the dimension-specific forces is non-trivial, then all forces may be accounted for in the physical input force vector.

(168) A frictional input force vector is determined at **2010**. In some embodiments, the friction forces may be applied in the direction opposite to velocity in each dimension. For instance, if the robot is rotating clockwise and moving in a forward direction, then the friction force input vector may be vector with values applying force in an anticlockwise and backward direction. The current direction of movement may be determined based on the operating conditions determined at **2006**.

(169) According to various embodiments, a frictional input force vector may be composed of various components. The frictional input force vector may be determined by aggregating these components, for instance by summing them. Examples of components that may be included in a frictional input force vector include, but are not limited to: Coulomb friction, damping friction, static friction, dynamic friction, other types of friction, and/or combinations thereof.

(170) In some embodiments, as discussed above, a Coulomb friction component may be applied. For instance, the Coulomb friction force vector may be constant in the direction against motion. The Coulomb friction may, for instance, cause the robot to slow down over time absent user input. The particular values used for Coulomb friction may depend on a variety of factors, such as the weight of the robot. For instance, a Coulomb friction coefficient between 0.01 and 0.25 may be applied by multiplying the coefficient by the weight of the robot.

(171) In some embodiments, as discussed above, a damping friction component may be applied. For instance, the damping friction vector may be proportional to velocity in the direction against motion. The damping friction vector may, for example, limit the top speed of the robot in any direction. The damping friction vector may also help to reduce stability concerns, for instance in the event that a robot impacts an obstacle and then rebounds sharply in the opposite direction. In some configurations, a damping friction coefficient between 0.5 and 2.0 may be applied by multiplying the coefficient by the weight of the robot.

(172) A functional input force vector is determined at **2012**. According to various embodiments, the functional input force vector may be used to supply force based on functional considerations such as safety, obstacle avoidance, and/or other operational goals. In some embodiments, the functional

input force vector may be used to guide the robot to a particular location or along a particular route within an environment. For example, the functional input force vector may include a virtual magnet force that pulls the robot along a route or to a designated location provided as input. For instance, the functional input force vector may be used to move the robot along a route through an environment, effectively guiding the operator along. However, the operator may be able to override the functional input force by, for instance, exerting a sufficiently strong physical force in a different direction.

(173) In some embodiments, the functional input force vector may be used to guide the robot safely within an environment. For example, to facilitate obstacle avoidance, the functional input force vector may include a virtual repellant force that causes walls, people, and/or other obstacles to effectively push back against the robot. As another example, the functional input force vector may include virtual rumble strips that vibrate the robot under one or more operating conditions.

(174) In some embodiments, the functional input force vector may be used to enforce a speed limit. For example, the functional input force vector may include a component that pushes back in a direction against velocity to prevent the robot from attaining a speed greater than designated maximum. The designated maximum speed limit may change depending on one or more considerations, such as the robot's location within an environment.

(175) In some embodiments, the functional input force vector may be used to provide virtual haptic rails or virtual train rails. Haptic rails may be modeled as a virtual track along which the robot tries to maintain alignment. Moving the robot off of the virtual corridor may require user input such as sharp torque to the handlebar to “pop” the robot off of the rail. At the moment of leaving the rail, the robot may apply a sharp impulse such as a pop or a step in velocity to simulate leaving a track.

(176) In some embodiments, haptic rails or virtual train rails may be defined in any of various ways. For example, the robot may project rails onto the ground via a projector. As another example, haptic rails, virtual train rails, and/or areas of particular speed zones may be detected by a robot based on, for instance, tape or paint applied to a region of the ground.

(177) In some embodiments, the functional input force vector may be used to simulate dynamic rails that lead the operator in a particular direction. For instance, the operator may be guided to move the robot along a virtual track, with movement along the track requiring much less force to the handlebars than movement in a different direction. The rails may be sharp or soft, and may be narrow or wide, depending on the application.

(178) According to various embodiments, the location of virtual rails or obstacles or the initialization of a strafing mode may be determined in various ways. For example, environment detection may involve input from a visual SLAM, inertial measurement unit, or other such data source. As another example, such a mode may be detected based on user input or one or more configuration parameters. For instance, the robot may automatically enter a strafing mode when it enters an aisle. As still another example, virtual rails may be created based on lines painted or projected on the floor. Alternatively, the robot may project rails onto the floor that match the virtual rails being used by the robot.

(179) In some embodiments, the functional input force vector may be used to assist in smooth obstacle avoidance. For instance, the robot may use the functional input force vector to simulate a repelling and/or dampening force when approaching an obstacle, finally slowing to a stop without hitting the obstacle and despite user input on the handlebars pushing the robot in the direction of the obstacle. The functional input force vector may be used to introduce jitter simulating a rumble strip when moving toward an obstacle or other area where the robot determines that it should not travel.

(180) In some embodiments, the functional input force vector may be used to provide haptic feedback via one or more motors in a drive unit on the robot. The strength of the haptic feedback and/or the force needed to operate the robot may be adapted for the individual.

(181) The identity of the individual may be determined based on an identifier such as a badge, a bar

code, an RFID tag, or other such indicator. Alternatively, or additionally, the adaptation may rely on an estimation of strength as a function of the user's size or force exerted on the robot.

(182) In some embodiments, the functional input force vector may be used to model the inertia of the device and then change that inertia as experienced by the user. Such a model may be used, for instance, to make the robot seem lighter in rotation than an unpowered robot would be given its mass, or to move the effective center of rotational mass backward toward the operator to simulate a “shopping cart” feeling. As still another example, the functional input force vector may be used to lock the orientation of the robot into a “strafing mode”. In the strafing mode, the robot may align its orientation with an aisle, grid, user, or other reference point. A strafing mode may be used to simulate inertia based on a direction. For instance, the robot may snap to a virtual grid but effectively provide a preference for a longitudinal axis. In such a configuration, the strafing mode may help to avoid drift down a long straightaway. Similarly, the robot may seem to be lighter in the preferred direction, with motion in a different direction requiring higher activation force.

(183) In some embodiments, the functional input force vector may include a component damping motion in a direction lateral to the front of the robot. For example, such a component may be used, for instance, to facilitate smooth motion in the direction intended by the operator. For instance, such a force may facilitate smooth turning by increasing the force in the direction in which the robot is pointing.

(184) In some embodiments, the functional input force vector may include multiple components. For example, the functional input force vector may include a combination of functional input force vector components corresponding with (1) haptic rails, (2) a speed limit, (3) a navigation objective, and/or any other elements.

(185) In some embodiments, one or more components of a functional input force vector may be determined subject to one or more constraints. For example, an obstacle avoidance force component, a speed limit force, and/or other such forces may be limited to operating in a direction opposed to velocity.

(186) In some embodiments, an unforced sensing mode may allow the robot to be moved without force detection. However, moving a heavy omnidirectional object without activating a drive mechanism can be difficult due to the challenge in changing inertia, such as when turning a corner. To address such a challenge, the functional input force vector may be used to simulate virtual fixed wheels at a point in the robot, making the robot more easily turnable in the unforced sensing model. In some configurations, the location of the virtual fixed wheels may be configurable or adjusted automatically, for instance being moved based on physical slide being detected.

(187) In some embodiments, the functional input force vector may be applied in an unforced sensing mode for safety and/or convenience purposes. For instance, forward force may be applied to compensate for friction. As another example, stopping force may be applied whenever acceleration over a given threshold is detected.

(188) In some embodiments, a mobile robot may lock to orientation, for instance in strafing mode or when following an operator. When locking orientation, the robot may resist attempts to rotate the robot. For instance, applying force to one corner of the robot in a way that would normally cause the robot to rotate may lead to the robot applying a virtual force to another corner to maintain the orientation. Such locking may occur even in an unforced sensing mode, for instance by measuring displacement, wheel movement, inertia, and/or other such sensor values.

(189) In some embodiments, the functional input force vector may depend at least in part on configuration settings that may be adapted to the user. For example, a user may “level up” with experience to allow the use of additional features. As another example, some features may be disabled, depending on the application or area in which the robot is operating. As still another example, an operator or fleet manager may activate or disable one or more features.

(190) In some implementations, the functional input force vector may include one or more elements received from a remote computing device, such as a fleet controller. The one or more

elements may include force vector components to be included in the functional input force vector, a function to calculate such components, a goal to be used in calculating such components, and/or other suitable information.

(191) One or more robot drive unit control output instructions are determined at **2014** based on the one or more movement control input instructions. In some embodiments, a movement control output instruction may be provided to the robot drive unit to cause the robot drive unit to apply force to the robot. The movement control output instruction may be determined by combining different types of movement control input information into an instruction capable of being acted upon by the robot drive unit.

(192) According to various embodiments, a movement control output instruction may be specified as a vector in one or more dimensions. For instance, a vector may include different directional components corresponding to movement in the x-direction, the y-direction, and the rotational direction with the robot being positioned on a virtual x-y plane corresponding with the physical environment in which the robot is situated. A directional component may indicate a magnitude associated with movement in the indicated dimension. The magnitude may be indicated as, for instance, a value corresponding with a velocity or a force.

(193) Various techniques may be used to determine the one or more robot drive unit control output instructions. Additional details regarding such techniques are discussed with respect to the method **2200** shown in FIG. 22.

(194) The one or more robot drive unit control output instructions are provided to a drive unit for the robot at **2016**. In some embodiments, the robot drive unit control output instructions may be provided to an abstraction layer for the robot, such as the abstraction layer **2108** shown in FIG. 21. The abstraction layer may provide separation between: (1) the determination of how the robot should move; and (2) the control of the hardware components of the drive unit to achieve that movement. In this way, the same controller logic may be applied to different physical configurations of drive units.

(195) As shown in FIG. 21, the robot drive unit abstraction layer **2108** includes a control interface **2112**, an instruction translator **2114**, and a drive interface **2116**. The control interface **2112** receives the control instructions from the robot controller **2106**. The instruction translator **2114** translates those control instructions into hardware-level instructions executable by the robot drive unit **2110**. The drive interface **2116** then provides those control instructions to the robot drive unit **2110**. For instance, the robot drive controller **2106** may issue a control output instruction to the robot drive unit abstraction layer **2108** through the control interface **2112** effectively instructing the drive unit to move with force magnitude (f1, f2, f3) in the (x, y, r) direction. The instruction translator **2114** may then translate that control output instruction into individual motor-level instructions to one or more motors included in the drive unit **2110**, such as motors corresponding to different wheels.

(196) Returning to FIG. 20, a determination is made at **2018** as to whether to continue to control the robot drive unit. According to various embodiments, the robot drive unit may continue to be controlled until a terminating condition is met. For example, control may be terminated when the robot enters a deactivated state, an error condition, a charging mode, or another situation in which movement is not indicated.

(197) According to various embodiments, the operations **2004** through **2018** may be performed with any suitable frequency. For instance, this control loop may operate at a rate between the range of 25 hertz to 2 kilohertz.

(198) FIG. 22 illustrates a method **2200** for determining a robot control output instruction, performed in accordance with one or more embodiments. The method **2200** may be performed at the robot drive controller **2106** shown in FIG. 21. The robot control output instruction may be specified in accordance with the control interface **2112** of the robot drive unit abstraction layer **2108**. For instance, the robot control output instruction may be specified as a vector identifying values corresponding to magnitude of either force or velocity in one or more dimensions.

(199) A request to determine one or more robot control output instructions is received at **2202**. In some embodiments, the request may be generated in the course of executing a robot control loop. For instance, the request may be generated as discussed with respect to operation **2014** shown in FIG. **20**.

(200) A determination is made at **2204** as to whether to provide velocity-based control instructions. In some embodiments, the robot may be controlled by directing the drive unit to move the robot in accordance with a vector that specifies velocity values along one or more dimensions.

Alternatively, the robot may be controlled by directing the drive unit to move the robot in accordance with a vector that specifies force values along one or more dimensions.

(201) In some embodiments, the robot may operate entirely in one mode or another. For instance, the robot may be configured to operate entirely in a force-based control mode or a velocity-based control mode. Alternatively, the mode may be dynamically determined based on one or more considerations such as user input, operating conditions, payload weight, location, and/or communication with a remote computing system such as a fleet controller.

(202) According to various embodiments, the determination made at **2204** may reflect one or more tradeoffs. For instance, employing velocity-based controls for a robot may allow the robot to operate with a constant apparent mass to the user, regardless of the actual mass of the robot and any load carried by the robot. However, a velocity-based control approach may create a counterintuitive situation in which the forces exerted on the robot that are not detected by the force sensor are effectively counteracted through the application of the velocity-based control logic since the robot drive unit is controlled so as to match the target velocity. Employing instead a force-based control through the use of a force multiplier may allow the robot to take into account forces exerted on the robot that do not act through the force sensor. However, loading the robot with an increased cargo mass will result in the operator feeling the additional mass in the absence of dynamic adjustment to the force multiplier, discussed in greater detail below.

(203) Upon determining to apply a force-based control instruction, a force multiplier is determined at **2206**. In some embodiments, the force multiplier may include one or more values applied as a multiplier to the force sensor force vector to determine control instructions for the robot. In this way, a user may move the robot in a particular direction by applying a force potentially much smaller than what would be required to move the robot in the direction were no force assistance provided.

(204) In some embodiments, the force multiplier may be implemented as a scalar. In such a configuration, the same force multiplier may be applied to all dimensions. Alternatively, the force multiplier may be implemented as a vector. In such a configuration, different dimensions may receive different force multiplication. For example, the rotational dimension may be associated with a larger force multiplier due to the difficulty (relative to translational force) of applying rotational force to an input device such as a handlebar.

(205) In some embodiments, the force multiplier may be fixed. For instance, a force multiplier of 1.5 $\times$, 2 $\times$, 3 $\times$, or another suitable value may be used. Alternatively, the force multiplier may be dynamically determined. For instance, the force multiplier may be increased or decreased based on the operating conditions optionally determined at **2006**. In some configurations, the force multiplier may be determined based at least in part on configuration information provided by a fleet administrator and/or robot user.

(206) In some embodiments, a force multiplier may be increased relative to a previously used and/or default value. As an example, a lower force multiplier may be used: (1) when the robot is located in an area designated for slower speed, (2) when a potentially unsafe condition is detected, (3) when a larger or stronger operator is using the robot, (4) when the robot has been manually configured to have a lower force multiplier, (5) when the robot has been detected as carrying less mass, and/or (6) when any other type of operating condition designated for use with a lower force multiplier is detected.

(207) In some embodiments, a force multiplier may be increased relative to a previously used and/or default value. For example, a higher force multiplier may be used (1) when the robot is located in an area designated for higher speed, (2) when it is determined that the robot is operating in an area of low traffic, (3) when the robot has been manually configured to have a higher force multiplier, (4) when the robot has been detected as carrying more mass, when a smaller or weaker operator is using the robot, and/or (5) when any other type of operating condition designated for use with a higher force multiplier is detected.

(208) In some embodiments, a force multiplier may depend on the direction of the physical force relative to a current velocity of the robot. For example, a higher force multiplier may be used when the physical force is in the opposite direction as the velocity. In this way, an operator may be able to slow or stop the robot more easily than the operator can increase the robot's speed. In some configurations, such determinations may be made on a dimension-by-dimension basis. For instance, in a situation in which the robot is moving in the x-direction while rotating counterclockwise, a physical force applied clockwise and in the x-direction may receive a higher force multiplier in the rotational direction than in the translational direction since the rotational physical force is against the direction of velocity while the translational force in the same direction as the velocity.

(209) In some embodiments, a force multiplier may depend on the location of the robot relative to the operator. For example, a larger force multiplier may be used when moving the robot in a direction away from the operator than when moving the robot in a direction toward the operator. In this way, the robot may be safely accelerated away from the operator while at the same time preventing the operator from inadvertently cause the robot to unsafely accelerate toward the operator.

(210) An updated physical input force vector is determined at **2208**. In some embodiments, the physical input force vector may be determined by multiplying the physical input force vector determined at **2008** by the force multiplier determined at **2206**.

(211) An output force vector is determined at **2210** based on the input force vectors. In some embodiments, the output force vector may be determined by combining the updated physical input force vector determined at **2206** (i.e., $k \cdot \text{Math.F.sub.sensed}$, where k is the force multiplier and F.sub.sensed is the force detected at the force sensor(s)) with the frictional input force vector determined at **2210** (i.e., F.sub.friction) and the functional input force vector determined at **2212** (i.e., F.sub.functional). For instance, the various input force vectors may be summed to determine the output force vector. That is, $\text{F}_{\text{applied}} = k \cdot \text{Math.F.sub.sensed} + \text{F.sub.friction} + \text{F.sub.functional}$.

(212) Upon determining instead to provide velocity-based control instructions, a virtual mass value for the robot is identified at **2212**. By causing the robot to move at a velocity consistent with the virtual mass value when acted upon by a force applied to the robot by a user, the virtual mass value may be used to cause the robot to feel as if it weighs a particular amount.

(213) In some embodiments, the virtual mass may be specified as a configuration value. In some configurations, the virtual mass may be fixed. For instance, a robot may be assigned a virtual mass of 60 pounds. Alternatively, the virtual mass may be dynamically determined. For example, the virtual mass value may be increased for users identified as being relatively larger and/or stronger. As another example, the virtual mass value may be dynamically determined based on observations about user input over time.

(214) An acceleration vector for the robot is determined at **2214** based on the force vector and the virtual mass. In some embodiments, the acceleration vector may be determined by dividing the force vector by the virtual mass. Under Newton's second law of motion, the acceleration of an object is equal to the force applied to the object divided by the object's mass. Thus,

$$(215) a = \frac{F_{\text{sensed}} + F_{\text{friction}} + F_{\text{functional}}}{m}$$

where m is the virtual mass and a is the acceleration vector.

(216) A velocity output vector for the drive unit is determined at **2216**. In some embodiments, the

velocity output vector may be determined based on the formula $v_{sub.t} = v_{sub.t-1} + a \cdot \Delta t$, where $v_{sub.t-1}$ is the current velocity vector for the robot, a is the acceleration vector for the robot, and Δt is the time interval for one iteration of the drive unit control loop. Thus, the velocity output vector may be obtained by integrating the acceleration vector over a suitable period of time.

(217) In some embodiments, the current velocity vector may be identified as a vector in dimensions corresponding to those of the acceleration vector determined at **2214**. The current velocity vector may be determined as part of the operating conditions determined at **1606**. For instance, one or more sensors associated with the drive unit may provide sensor data indicating the velocity at which the robot is currently traveling.

(218) FIG. **23** illustrates a method **2300** for autonomous motion control of an autonomous mobile robot, performed in accordance with one or more embodiments. The method **2300** may be used to direct an autonomous mobile robot to perform a task, for instance by aiding an operator.

(219) A request to autonomously control an autonomous mobile robot in a physical environment is received at **2302**. In some embodiments, the request may be received when the robot enters an autonomous mode, as discussed with respect to operation **1904**.

(220) A scene graph of the physical environment is determined at **2304**. In some embodiments, the scene graph may provide a virtual representation of a physical environment. For instance, a scene graph of a warehouse may provide a virtual representation of aisles in the warehouse, along with locations of items, zones for dropping off items, and the like. The scene graph may be connected to one or more data sources, for instance allowing the robot to determine a correspondence between an item to be picked up or dropped off and a location in the physical environment. For example, a database may indicate that Item A134 is located in Bin 23 on Shelf 10 of Aisle 9, and the scene graph may provide a virtual representation location of Bin 23, Shelf 10, Aisle 9 in a way that allows the robot to navigate to that location.

(221) In some embodiments, the scene graph may be received from a remote computing device. For instance, the scene graph may be received from a fleet controller configured to control multiple robots. Alternatively, or additionally, elements of the scene graph may be determined by the robot itself. For instance, the robot may analyze sensor data to determine or supplement a scene graph. As still another possibility, a scene graph may be received from a remote computing device and then updated by the robot.

(222) A current location for the robot on the scene graph is determined at **2306**. According to various embodiments, the location of the robot may be determined in any of various ways. For example, image data from one or more cameras at the robot may be analyzed using a visual SLAM and/or other techniques to determine a location of the robot relative to one or more reference points. A correspondence between a reference point and the scene graph may then allow the robot to determine its location on the scene graph.

(223) A task to perform is determined at **2308**. A destination location on the scene graph is determined at **2308** based on the task. A route from the current location to the destination location is determined at **2310** based on the scene graph. The robot is instructed to move at **2310** based on the movement instruction. According to various embodiments, the particular task, location, and movement instruction may depend in significant part on the application.

(224) In some embodiments, the robotic cart may be configured to aid in the completion of a task in a warehouse. For example, the robot may be configured to aid in a task such as item picking in which it retrieves one or more items from storage in the warehouse for transport to another location. As another example, the robot may be configured to aid in a task such as item replenishment in which it delivers one or more items to one or more locations within the warehouse for future picking. As still another example, the robot may be configured to aid in another task, such as transporting a person, transporting a production input, transporting a production output, providing a mobile light source, monitoring a region, monitoring a person, removing trash, or the like.

(225) In some embodiments, item picking may be performed using any of a variety of protocols. For example, in zone picking, a person may operate in an area of the warehouse to pick items while one or more robots travel to the person to collect those items. As another example, in batch picking, the robot may be equipped with one or more boxes, some or all of which may include items corresponding with multiple orders. As still another example, an autonomous mobile robot may be configured to follow a person as the person moves around a warehouse environment and places items into the robotic cart. As yet another example, an autonomous mobile robot may be configured to facilitate order consolidation, in which it moves to a location close to another robot and supports the movement of items between the two carts.

(226) In some embodiments, in support of these tasks, the robot may use lights or other affordances to interact with humans. For example, a light strip may include lights that may be adapted to a width of a region on a cart. The lights may then be activated to indicate to a human where to remove an item from or where to place an item on the cart.

(227) In some embodiments, a task for the robot to perform may be determined by a coordinator such as a fleet management system providing command and control instructions for a fleet of robots. In such a configuration, the fleet management system may perform route determination and/or optimization. For example, the fleet management system may spread the load to avoid traffic congestion, determine estimates for task completion time, and/or generally determine tasks in a manner that efficiently utilizes multiple robots.

(228) In some embodiments, the route may be parameterized as a nominal trajectory that includes one or more individual waypoints. Each individual waypoint may be a location within the scene graph. However, the robot need not necessarily follow the exact path specified by the nominal trajectory. Instead, the path may be treated as, for instance, a force vector that serves as an input along with other force vectors, such as those associated with obstacle avoidance, that are combined together to determine a direction and magnitude of movement for the robot at any given time.

(229) A determination is made at **2312** as to whether to continue controlling the robot. According to various embodiments, the robot may continue to be controlled as long as it remains in autonomous motion mode.

(230) In some embodiments, an autonomous mobile robot may be equipped with one or more sensors such as visible light cameras. FIG. **24** illustrates a diagram **2400** showing sensor placement on an autonomous mobile robot. Depending on the robot configuration, sensors may be located in various places. Locating a sensor lower on the autonomous mobile robot, for instance as shown at the camera **2404** in FIG. **24**, provides improved obstacle detection for small obstacles located on the floor due to reduced depth noise. In contrast, locating a sensor higher on the autonomous mobile robot, for instance as shown at the camera **2402** in FIG. **24**, provides for improved visibility of people close to the autonomous mobile robot and can avoid creating a blind spot around the base. However, such a configuration can reduce the field of view around the robot and render small objects located on the floor more difficult to detect.

(231) Robot Light Control Processes and Architecture

(232) FIG. **25A** illustrates a view of an attachment point **2502** between a lighting element **2506** and a communication channel **2504** in an angled shelf configuration, while FIG. **25B** illustrates a view of an attachment point **2512** between a lighting element **2506** and a communication channel **2514** in a flat shelf configuration. According to various embodiments, the lighting elements **2506** and/or **2516** may be coupled with the communication channels **2504** or **2514** and may provide lighting in a way that assists in the performance of one or more operations using the autonomous mobile robot. For example, the lighting elements **2506** and/or **2516** may include one or more pick-to-light components that highlight one or more external locations such as a location on a shelf in the physical environment in which the autonomous mobile robot is situated. As another example, the lighting elements **2506** and/or **2516** may include one or more put-to-light components that highlight one or more internal locations on the autonomous mobile robot, such as a tote or shelf

region. Such lighting elements may facilitate workflows such as picking items from an external shelf and placing them onto the autonomous mobile robot, placing items from the autonomous mobile robot onto an external location such as a shelf or other robot, or the like.

(233) In some embodiments, lights may be used to support a virtual and/or regular pick wall and/or put wall. For example, a regular pick wall or put wall may be a wall of cubbies with lights on the robot and/or the cubbies indicating where to pick and/or place an item. As another example, a robot may be configured to operate as a virtual pick wall or put wall at any location by illuminating lights associated with locations on the robot's chassis (e.g., bins on shelves). In particular embodiments, multiple robots may coordinate to support item transfer between and among the robots and/or another location such as a regular pick and/or put wall.

(234) FIG. **26** illustrates a method **2600** for controlling one or more lights associated with an autonomous mobile robot, performed in accordance with one or more embodiments. A request to control a light associated with an autonomous mobile robot is received at **2602**. A task being performed by the robot or an operator is identified at **2604**. A light configuration in support of the task is determined at **2606**. One or more lights are activated based on the light configuration at **2608**. A determination is made at **2610** as to whether to continue to control a light associated with the robot.

(235) According to various embodiments, the method **2600** may be used to configure various types of lights. For example, the robot may be equipped with projectors that can project light onto a surface inside the robot, such as onto a shelf location. For example, the robot may be equipped with projectors that can project light onto a surface off of the robot, such as onto a bin located on a shelf in a warehouse aisle. As still another example, the robot may be equipped with a light strip that can highlight an area of an autonomous mobile robot, such as a region of a shelf. A light may be configured as a laser beam on a gimbal, a laser line, and addressable LED strip, a fixed light whereby the robot moves the light by aligning the robot with the shelf, and/or any other suitable lighting configuration.

(236) According to various embodiments, the method **2600** may be used to configure various types of lighting operations. For example, the robot may be configured to light up an area or container corresponding with where an item is to be picked up from or moved to. For instance, the robot may project a light onto an external bin corresponding to an item to be picked up and then instruct an addressable LED strip to light up an area on the robotic car where the item is to be placed.

(237) Navigational Feedback

(238) FIG. **27** illustrates a navigational feedback method **2700**, performed in accordance with one or more embodiments. The navigational feedback method **2700** may be performed at an autonomous mobile robot configured in accordance with techniques and mechanisms described herein.

(239) A request to provide navigational feedback for an autonomous mobile robot is received at **2702**. In some embodiments, the request may be generated as discussed with respect to the operation **2012** shown in FIG. **20**. That is, navigational feedback may be generated in the context of the execution of a robot control loop.

(240) In some embodiments, navigational feedback may include a haptic feedback force input vector, which may be a type of functional input force vector that may be provided to the mechanical drive unit. Alternatively, or additionally, navigational feedback may include other types of feedback such as visual feedback or auditory feedback.

(241) User input for the autonomous mobile robot is identified at **2704**. In some embodiments, the user input may include force detected at one or more force sensors associated with a handlebar unit. For instance, the user input may include one or both of a translational force and a rotational force exerted on the handlebar.

(242) One or more operating conditions associated with the autonomous mobile robot are identified at **2706**. According to various embodiments, as discussed with respect to the operation **2006**,

operating condition for a robot may include any of a variety of conditions.

(243) Examples of such operating conditions may include, but are not limited to: the identity of an operator using the robot, a location of the robot, a direction in which the robot is traveling, an amount of traffic in the vicinity of the robot, a condition associated with the physical environment in which the robot is situated, and the like.

(244) In some embodiments, the one or more operating conditions may include one or more situational aspects of the robot relative to its environment. Examples of such operating conditions may include, but are not limited to: a determination that the robot is moving along an aisle, a determination that the robot is approaching a real or virtual obstacle or barrier, a determination that the robot is approaching a zone associated with a different mode of operation, a determination that the robot is approaching the end of an aisle, a proximity of the robot to an edge of a virtual corridor, and/or other such considerations.

(245) In some embodiments, one or more operating conditions may include one or more functional aspects related to a task being performed by the robot, for instance in the context of a workflow. Examples of such operating conditions may include, but are not limited to: determining that a robot is approaching an area corresponding to an item to be moved, determining that a robot is moving away from an area corresponding to an item to be moved, determining that a human operator of the robot is to be notified of information, and/or other operational information.

(246) A determination is made at **2708** as to whether the one or more operating conditions and/or the user input meet one or more triggering conditions for providing navigational feedback. In some embodiments, the determination may involve evaluating whether the robot's position coupled with the user input suggests that the robot, absent a correction, is projected to cross a threshold, violate a rule, create a dangerous situation, and/or perform an action that is deemed inefficient. For example, the robot's position, direction, and user input may be collectively analyzed to project that the robot is heading toward a virtual barrier.

(247) In some embodiments, the determination made at **2708** may involve evaluating whether the robot's position coupled with the user input suggests that the robot is approaching or departing a virtual navigational affordance. For instance, a warehouse environment may be configured with one or more virtual high-speed navigational corridors, virtual low speed handling areas, virtual navigational rails, virtual parking areas, or the like. A robot approaching or departing from such a virtual navigational affordance may generate a haptic feedback pattern appropriate for the affordance, such as snapping into or out of a parking spot, navigational rail, or corridor.

(248) In some embodiments, the determination made at **2708** may involve evaluating whether the robot's position coupled with the user input suggests that the robot is approaching or departing a configuration associated with a workflow. For example, a consolidation workflow may involve an arrangement of two or more robotic carts in which the carts are positioned near each other to facilitate the transfer of items from one cart to another. Such a consolidation workflow may be associated with a predetermined relative spatial orientation of robotic carts to facilitate efficiency on the part of the human operator. As another example, a virtual train workflow may involve a sequence of carts configured to closely follow one another in a row, with a human user capable of creating such a sequence by moving the carts in proximity to one another. As with a virtual navigational affordance, a robot approaching or departing from a workflow configuration may generate a haptic feedback pattern appropriate for the affordance, such as snapping into or out of position.

(249) In some embodiments, the determination made at **2708** may involve evaluating whether the robot's position coupled with the user input suggests that the robot is entering or leaving a particular operating mode. For instance, in a guided navigation mode, the robot may autonomously navigate along a path despite the user's hand on the handlebar. In contrast, in a fully manual mode, the robot may cease autonomous navigation apart from obstacle avoidance and instead be fully responsive to user input. The robot may generate haptic or non-haptic feedback to indicate that

such a transition has occurred.

(250) In some embodiments, the determination made at **2708** may be made based at least in part on sensor data identifying one or more indicators of a virtual barrier, corridor, rail, zone, or other such navigational affordance. For example, paint or tape on the floor of a warehouse may be used to indicate the presence of a navigational affordance. As another example, a caution sign or other such warning may indicate the presence of a zone associated with reduced speed. As yet another example, a virtual navigational rail or corridor may be automatically created along a linear surface such as a shelf in a warehouse.

(251) Upon determining that the one or more operating conditions and/or the user input meet one or more triggering conditions for providing navigational feedback, non-haptic navigational feedback is determined and provided at **2710**.

(252) In some embodiments, non-haptic navigational feedback may include visual feedback provided via one or more light projectors. For example, the autonomous mobile robot may project onto the ground an indication of one or more virtual rails, barriers, or corridors. As another example, the autonomous mobile robot may project onto the ground or another surface an indication of a recommended direction or route of travel. As yet another example, the autonomous mobile robot may project onto the ground or another surface one or more messages or indicators.

(253) In some embodiments, non-haptic navigational feedback may include visual feedback provided via a display screen or light strip. For example, the autonomous mobile robot may update the display screen to display a map with a route, one or more turn-by-turn direction, or a message.

(254) In some implementations, non-haptic navigational feedback may include one or more sounds. For example, the autonomous mobile robot may play a sound to indicate that the user is guiding the autonomous mobile robot in a direction that is not recommended.

(255) Upon determining that the one or more operating conditions and/or the user input meet one or more triggering conditions for providing navigational feedback, a haptic feedback input force vector is determined and applied at **2712**. According to various embodiments, the haptic feedback force vector may be applied by combining it with other input force vectors as discussed with respect to the method **2000** to determine aggregate robot drive unit control output instructions for controlling the robot's mechanical drive unit.

(256) In some embodiments, the haptic feedback may be experienced by a human operator as a force sensation in the operator's hand while holding the force-sensitive handlebar. For instance, the drive unit may cause the entire autonomous mobile robot to jerk in a particular direction, vibrate, create a feeling of push or pull, or otherwise generate the feeling of force in the handlebar.

(257) According to various embodiments, one or more of various types of haptic feedback may be provided. In some configurations, haptic feedback may include generating a force vector that provides the feel of a bump in one direction. For example, such a pattern of haptic feedback may involve generating a short but sharp increase in force in one direction.

(258) In some embodiments, haptic feedback may include generating a force vector that provides a steady force in one direction or a force that steadily increases or decreases in intensity.

(259) In some implementations, haptic feedback may include a force vector generated so as to create via the drive unit a feeling of vibration, for instance by generating a temporal sequence of oscillating haptic feedback input force vectors pointed in opposite directions that collectively cancel out in terms of their ultimate effect on the directional motion of the autonomous mobile robot.

(260) In some embodiments, haptic feedback may include one or more complex patterns. For example, when it is determined that the autonomous mobile robot has nearly aligned with a virtual rail, a profile of haptic feedback vectors may be determined to provide the feeling that the robot has snapped into a physical rail. As another example, when it is determined that the autonomous mobile robot is moving along a virtual rail, the force vectors may initially act to keep the autonomous mobile robot aligned with the rail, effectively resisting or ignoring user input forces in the direction

orthogonal to the rail or in the rotational direction. However, if a sufficiently strong lateral or rotational user input force is detected, a profile of haptic feedback vectors may be determined to provide the feeling that the robot has snapped out of a physical rail.

(261) In some embodiments, the presence of haptic feedback may cause the autonomous mobile robot to temporarily nullify or ignore input provided via the handlebar. For example, the haptic feedback force input vector may cause the autonomous mobile robot to suddenly snap into or out of a virtual rail, temporarily disregarding user input to the contrary.

(262) According to various embodiments, the particular type of haptic feedback that is provided may be strategically determined based on factors such as the characteristics of the autonomous mobile robot, the physical environment in which it operates, the tasks for which it is employed, and the human users operating the autonomous mobile robot. However, additional examples of the types of haptic feedback that may be provided and the triggering conditions for generating the haptic feedback are discussed with respect to FIG. 28A through FIG. 30.

(263) In some embodiments, non-haptic navigational feedback may be provided in the absence of haptic navigational feedback, or vice versa. Further, non-haptic navigational feedback may be provided as a precursor to providing haptic navigational feedback. For instance, before causing the drive unit to vibrate the robot, the robot may first display or project blinking lights, emit a sound, or provide other such non-haptic feedback. Alternatively, non-haptic feedback may be provided during or after haptic feedback is provided. For instance, a display screen may be updated to display a recommended path of action after or during a time in which the robot provides haptic feedback via the mechanical drive unit.

(264) In some embodiments, the triggering conditions for providing navigational feedback may be configurable by an environment administrator, system administrator, and/or end user. Thus, the examples of triggering conditions and navigational feedback described herein are provided as illustrative examples and are not an exhaustive list of the types of triggering conditions and feedback that may be used. Rather, the triggering conditions and navigational feedback may be strategically determined based on factors such as the characteristics of the autonomous mobile robot, the physical environment in which it operates, the tasks for which it is employed, and the human users operating the autonomous mobile robot.

(265) FIG. 28A, FIG. 28B, and FIG. 28C illustrate diagrams of situations in which haptic feedback may be provided via a mechanical drive unit, generated in accordance with one or more embodiments. In FIG. 28A, an autonomous mobile robot **2800** is traveling obliquely toward a virtual rail **2802**. The autonomous mobile robot **2800** receives as input an input force vector **2804**. The input force vector **2804** includes a consistent force component **2806** in the direction of the virtual rail **2802** and an orthogonal force component **2808** that is orthogonal to the direction of travel. Because the autonomous mobile robot **2800** has not yet arrived at the virtual rail **2802**, the direction of travel **2810** mirrors the user input force vector **2804**. That is, haptic feedback is not provided.

(266) In FIG. 28B, the autonomous mobile robot **2800** has reached the virtual rail **2802**. The autonomous mobile robot **2800** receives as input an input force vector **2834**. The input force vector **2834** includes a consistent force component **2836** in the direction of the virtual rail **2832** and an orthogonal force component **2838** that is orthogonal to the direction of travel.

(267) In FIG. 28B, the consistent force component **2836** is relatively large, while the orthogonal force component **2838** is relatively small. In such a situation, an oppositional haptic feedback force input vector **2840** may be generated. Because the oppositional haptic feedback force input vector **2840** is directionally opposed to the orthogonal force component **2808** and of equal magnitude, the two force vectors cancel out each other. The resulting direction of travel **2842** is in the same direction as the virtual rail **2802**.

(268) Thus, the configuration shown in FIG. 28B initially provides the sensation of the autonomous mobile robot **2800** being snapped into the virtual rail **2802** when the autonomous mobile robot

2800 reaches the virtual rail **2802**. Such a situation may continue even if the hum orthogonal force component **2838** is maintained.

(269) After the autonomous mobile robot **2800** is snapped to the virtual rail, small forces in the orthogonal direction may be effectively ignored. Thus, the situation is one in which the human operator seems to be guiding the autonomous mobile robot **2800** along the virtual rail **2802** while, perhaps inadvertently, exerting a slight force orthogonal to the virtual rail **2802**. This approach helps the human user maintain the position of the autonomous mobile robot **2800** on the virtual rail **2802** without needing to steer the autonomous mobile robot **2800**.

(270) In FIG. **28C**, the autonomous mobile robot **2800** receives as input an input force vector **2854**. As in FIG. **28B**, the input force vector **2854** includes a consistent force component **2856** in the direction of the virtual rail **2852** and an orthogonal force component **2808** that is orthogonal to the virtual rail **2802**. However, in contrast to FIG. **28B**, the orthogonal force component **2858** is large.

(271) In such a situation, the autonomous mobile robot **2800** does not generate an oppositional haptic feedback force input vector, and instead may optionally generate an orthogonal haptic feedback force input vector **2860** in the direction of the orthogonal force component **2858**. The combination of these vectors leads to a direction of travel **2862** away from the virtual rail **2802** in the direction of the user input force vector **2854**.

(272) Thus, the configuration shown in FIG. **28C** provides the sensation of the autonomous mobile robot **2800** being snapped off of the virtual rail **2802** based on a large orthogonal force applied by the human user. This approach allows the human user to leave the virtual rail **2802** intentionally.

(273) The configurations shown in FIG. **28A**, FIG. **28B**, and FIG. **28C** involve static force vectors. However, changes in force vectors may also be considered when determining or applying haptic feedback. For instance, snapping the autonomous mobile robot **2800** off of the virtual rail **2802** may require a sharp increase in the orthogonal force component **2858** rather than steady pressure. That is, a sharp jerk of the autonomous mobile robot **2800** in the orthogonal direction may cause the autonomous mobile robot **2800** to suddenly leave the virtual rail **2802**, while a continuous force in the same direction may not have such an effect.

(274) FIG. **29A**, FIG. **29B**, and FIG. **29C** illustrate diagrams of situations in which haptic feedback may be provided via a mechanical drive unit, generated in accordance with one or more embodiments. In FIG. **29A**, the autonomous mobile robot is traveling in a virtual corridor between the virtual corridor wall **2902a** and the virtual corridor wall **2902b**. As the proximity **2904** of the autonomous mobile robot **2800** to the virtual corridor wall **2902b** decreases, an oppositional haptic feedback force input vector **2906** is applied to gently push the autonomous mobile robot **2800** back toward the center of the virtual corridor.

(275) In the configuration shown in FIG. **29A**, the human operating the autonomous mobile robot **2800** is in control of the autonomous mobile robot **2800**. However, the human operator receives guidance for keeping the autonomous mobile robot **2800** within the virtual corridor.

(276) In FIG. **29B**, the autonomous mobile robot **2800** has moved close to the virtual corridor wall **2902b**. In response to the decreased proximity **2914**, the oppositional haptic feedback force input vector **2916** is increased in magnitude, pushing the autonomous mobile robot **2800** more forcefully back toward the center of the virtual corridor.

(277) In the configuration shown in FIG. **29B**, the human operating the autonomous mobile robot **2800** remains in control of the autonomous mobile robot **2800**. However, the human operator receives more forceful guidance for keeping the autonomous mobile robot **2800** within the virtual corridor.

(278) In FIG. **29C**, the autonomous mobile robot **2800** has moved over the virtual corridor wall **2902b**. In response to such a traversal coupled with the close proximity **2924**, the orthogonal haptic feedback force input vector **2916** is provided, pushing the autonomous mobile robot **2800** away from the virtual corridor.

(279) In the configuration shown in FIG. **29B**, the human operating the autonomous mobile robot

2800 again remains in control of the lateral steering of the autonomous mobile robot **2800**.

However, the human operator receives guidance to remove the autonomous mobile robot **2800** entirely from the virtual corridor rather than operating it partially within the virtual corridor. Such guidance may help to keep the virtual corridor clear for other robots or humans.

(280) In some embodiments, a virtual corridor may be dynamically determined based on the routes planned for multiple robots. For instance, when multiple robots are moving in the same direction, they may coordinate to adhere to a dynamically determined virtual corridor in that direction. Then, the robots may generate navigational guidance through haptic feedback to stay within the virtual corridor. If a robot departs from the virtual corridor it may also generate navigational guidance through haptic feedback to fully depart the virtual corridor. In this way, the other robots traveling in the virtual corridor need not adjust their courses.

(281) FIG. **30A** and FIG. **30B** illustrate diagrams of situations in which haptic feedback may be provided via a mechanical drive unit, generated in accordance with one or more embodiments. In FIG. **30A**, the autonomous mobile robot **2800** is approaching a virtual barrier **3002** based on a user input force vector **3004**. As the autonomous mobile robot **2800** approaches the virtual barrier **3002**, the proximity **3006** is decreased. This decrease in the proximity **3006** triggers the generation of a haptic feedback force input vector sequence **3008**.

(282) In some embodiments, the haptic feedback force input vector sequence **3008** may be implemented as a sequence of individual haptic feedback force input vectors over time **3010**. For instance, the haptic feedback force input vector sequence **3008** may include a sequence of individual force vectors pointed in opposite directions.

(283) In some embodiments, the collective effect of the haptic feedback force input vector sequence **3008** may be experienced by the human operator as a back-and-forth vibration in the autonomous mobile robot **2800**. The vibration may be felt in the handlebar attached to the force sensor, or any other portion of the autonomous mobile robot **2800** in contact with the human operator.

(284) In FIG. **30B**, the autonomous mobile robot **2800** is participating in a task that involves navigating to a target **3054**. However, the autonomous mobile robot **2800** has passed the target **3054**, responsive to the user input force vector **3054**. As the proximity **3056** to the target **3054** decreases, the autonomous mobile robot **2800** generates a vibrational haptic feedback force input vector sequence **3058**.

(285) According to various embodiments, the vibrational haptic feedback force input vector sequence **3058** is substantially similar to the haptic feedback force input vector sequence **3008**. However, the vibrational haptic feedback force input vector sequence **3058** includes oppositional vectors arranged in an orthogonal, side-to-side direction rather than a parallel, forward-and-back direction.

(286) The configuration shown in FIG. **28A**, FIG. **28B**, FIG. **28C**, FIG. **29A**, FIG. **29B**, FIG. **29C**, FIG. **30A**, and FIG. **30B** as well as other configurations using haptic feedback, may support smooth and intuitive operation of the autonomous mobile robot **2800**. For instance, many autonomous mobile robots may operate in an area while maintaining both safe distances from one another and sufficiently high movement speed.

(287) In some embodiments, the presence of haptic feedback in combination with virtual navigational affordances may allow system operators to reduce the distance between robots without generating obstacle avoidance procedures. For example, in the absence of a virtual navigational affordance, two robots may be prevented from passing within one foot of each other by way of obstacle avoidance force vectors. However, such a situation may be permitted if each of the two robots is locked to a respective navigational rail.

(288) In some embodiments, haptic feedback may be used for purposes other than navigational guidance. For example, a bump or vibration may be used to notify the human operator of a message, which the human operator may access on a display screen associated with the

autonomous mobile robot. As another example, a sequence of slowly oscillating force vectors may serve to notify the human operator of a dangerous condition or other situation. Various configurations are possible.

(289) It should be noted that haptic feedback can be provided using force vectors in a rotational direction, and/or can be triggered based on one or more conditions associated with rotational movement. However, for the purposes of clear exposition, the illustrative examples presented herein have employed translational force vectors and omitted rotational components.

(290) Friction

(291) According to various embodiments, techniques and mechanisms described herein may be used to adjust the apparent friction felt by an operator of an autonomous mobile robot. FIG. **31** illustrates a method **3100** of determining a virtual friction vector, performed in accordance with one or more embodiments.

(292) A request to determine a virtual friction vector for an autonomous mobile robot is received. In some embodiments, the request may be received as discussed with respect to the operation **2010** shown in FIG. **20**.

(293) One or more operating conditions for the autonomous mobile robot are identified at **3104**. In some embodiments, the one or more operating conditions may be identified as discussed with respect to the operation **2006** shown in FIG. **20**.

(294) A physical friction vector for the autonomous mobile robot is determined at **3106**. In some embodiments, the physical friction vector may represent friction imposed by the components of the robot in one or more dimensions. Such friction may be due to components such as wheels, tires, motors, bearings, and the like. For instance, a robot may be anticipated to have a friction of between 1N and 100N based on such components.

(295) In some embodiments, the physical friction vector may be identified based on a predetermined configuration parameter. Alternatively, the physical friction vector may be dynamically determined based on a calibration method. An example of such a method is discussed with respect to the method **3200** shown in FIG. **32**.

(296) A desired friction vector is determined at **3108** based on the one or more operating conditions. According to various embodiments, various considerations may influence the configuration of the desired friction vector. The following examples are non-exhaustive.

(297) In some embodiments, lateral friction may be applied. Consider a robot that is moving along a straight line or a curve. Movement in a direction orthogonal to the line or curve may be referred to as the lateral direction. Friction may be applied in the lateral direction to oppose force applied to the robot to move the robot off of the line or the curve. In this way, the friction may help keep the robot on track and reduce the effort needed to control the robot. Such friction may potentially be increased as the robot's speed increases, for instance to provide additional stability.

(298) In some embodiments, forward friction force may be applied. Consider a robot that has is moving along a straight line or a curve, and suppose that the robot has a physical friction vector of 12N. Left uncorrected, the physical friction vector would represent a force that the operator would need to consistently apply to avoid having the robot slow down. Accordingly, a forward friction compensation vector of, for instance, 10N may be applied to correct for the robot's physical friction. To achieve such an effect, a desired friction vector of 2N in the backward direction may be specified. A small amount of desired friction may be maintained to avoid a situation in which the robot is felt to accelerate away from the operator.

(299) In some embodiments, a friction compensation force presented as forward friction may be applied when motion exceeds a designated velocity. That is, a dead band around zero velocity may be maintained. Alternatively, the friction compensation force may be gradually increased as velocity rises above zero according to a specified function.

(300) A virtual friction vector is determined at **3110** based on the desired friction vector and the physical friction vector. In some embodiments, the virtual friction vector may be determined by

subtracting the physical friction vector from the desired friction vector.

(301) FIG. **32** illustrates a method **3200** of calibrating a physical friction vector for an autonomous mobile robot, performed in accordance with one or more embodiments. In some embodiments, the method **3200** may be performed at the autonomous mobile robot. Alternatively, the method **3200** may be performed at a different location, such as a fleet controller. The method **3200** may be used to determine a physical friction vector employed as discussed with respect to the method **3100** shown in FIG. **31**.

(302) A request to determine a physical friction vector for an autonomous mobile robot is received at **3202**. In some embodiments, the request may be generated periodically or upon detection of a triggering condition. For instance, the request may be generated when a designated number of operating hours elapses.

(303) Input information for input vectors applied to the drive assembly is determined at **3204**. In some embodiments, the input information may include recent or current data that includes a set of observations for input vectors applied to the drive assembly over time. Such information may be stored at the robot itself and/or at a remote computing device.

(304) At **3206**, output information indicating motion of the autonomous mobile robot is determined. In some implementations, the output information may include values for velocity realized by the autonomous mobile robot in response to the input vectors identified at **3204**.

(305) A model of the autonomous mobile robot is determined at **3208** based on the input information, the output information, and a model specification. According to various embodiments, any of a variety of model specifications may be used. For instance, a polynomial model may represent values such as force, acceleration, velocity, mass, Coulomb friction, and the like. The model specification may then be applied to the input and output information and solved to determine an estimated friction vector.

(306) A physical friction vector is determined at **3210**. According to various embodiments, the physical friction vector may be determined based on the model determined at **3208**. For instance, the model determined at **3208** may include friction as a parameter that is solved for.

(307) In some embodiments, the friction vector may be represented as a single value. Alternatively, the friction vector may include multiple values, for instance corresponding to rotational and translational movement.

CONCLUSION

(308) In the foregoing specification, various techniques and mechanisms may have been described in singular form for clarity. However, it should be noted that some embodiments include multiple iterations of a technique or multiple instantiations of a mechanism unless otherwise noted. For example, a system uses a processor in a variety of contexts but can use multiple processors while remaining within the scope of the present disclosure unless otherwise noted. Similarly, various techniques and mechanisms may have been described as including a connection between two entities. However, a connection does not necessarily mean a direct, unimpeded connection, as a variety of other entities (e.g., bridges, controllers, gateways, etc.) may reside between the two entities.

(309) In the foregoing specification, reference was made in detail to specific embodiments including one or more of the best modes contemplated by the inventors. While various implementations have been described herein, it should be understood that they have been presented by way of example only, and not limitation. For example, some techniques and mechanisms are described herein in the context of industrial autonomous mobile robots configured for operation in a warehouse settings. However, the techniques of the present invention apply to a wide variety of autonomous mobile robots configured for operation in a wide variety of settings. Particular embodiments may be implemented without some or all of the specific details described herein. In other instances, well known process operations have not been described in detail in order not to unnecessarily obscure the present invention. Accordingly, the breadth and scope of the present

application should not be limited by any of the implementations described herein, but should be defined only in accordance with the claims and their equivalents.

Claims

1. An autonomous robotic cart comprising: a chassis including one or more shelves; a plurality of sensors coupled with the chassis, the plurality of sensors including a plurality of visible light cameras; a holonomic, omnidirectional, and backdriveable mechanical drive unit coupled with the chassis and being configured to cause the autonomous robotic cart to move along a planar surface; a handlebar unit coupled with the chassis, the handlebar unit including a handlebar and a force sensor configured to detect a translational force and a rotational force exerted on the handlebar, the handlebar being vertically positioned orthogonal to the planar surface; a processor configured to: transmit one or more instructions to the holonomic, omnidirectional, and backdriveable mechanical drive unit to autonomously navigate a physical environment to execute one or more navigation goals determined based on communication with a remote computing system configured to manage a fleet of robots including the autonomous robotic cart, determine an output force vector based on the translational force, the rotational force, sensor data received from a plurality of sensors, and a force multiplier applied to the translational force and the rotational force such that a translational force of interaction and a rotational force of interaction exerted by the handlebar falls below a designated threshold, transmit a second one or more instructions to the holonomic, omnidirectional, and backdriveable mechanical drive unit to cause the autonomous robotic cart to move translationally and rotationally in a direction corresponding to the output force vector; and a communication channel communicatively coupling the processor with the holonomic, omnidirectional, and backdriveable mechanical drive unit, the plurality of visible light cameras, and the force sensor.
2. The autonomous robotic cart recited in claim 1, wherein the processor is further configured to update a scene graph representing the physical environment based at least in part on the sensor data, and wherein the output force vector is determined based at least in part on an obstacle avoidance vector representing a virtual force on the autonomous robotic cart from a designated direction corresponding with an obstacle represented in the scene graph.
3. The autonomous robotic cart recited in claim 2, wherein the obstacle avoidance vector has a magnitude that increases in inverse proximity of the autonomous robotic cart with the obstacle.
4. The autonomous robotic cart recited in claim 2, wherein the processor is further configured to update the scene graph based at least in part on the communication with the remote computing system.
5. The autonomous robotic cart recited in claim 1, wherein the processor is further configured to detect a human within the physical environment based on the sensor data, and wherein the output force vector is determined based at least in part on a human avoidance vector representing a virtual force on the autonomous robotic cart from a designated direction corresponding with the human.
6. The autonomous robotic cart recited in claim 1, wherein the shelves are positioned in two or more layers, and wherein the chassis includes one or more vertical support elements supporting the layers, and wherein the handlebar is incorporated into a vertical support element of the one or more vertical support elements.
7. The autonomous robotic cart recited in claim 6, wherein the objects include one or more humans moving throughout the physical environment, and wherein the autonomous robotic cart is configured to identify the one or more humans as humans based on sensor data received from the plurality of sensors.
8. The autonomous robotic cart recited in claim 6, wherein the objects include one or more autonomous mobile robots moving throughout the physical environment, and wherein the autonomous mobile robot is configured to avoid the autonomous mobile robots based at least in

part on location information received from the remote computing system.

9. The autonomous robotic cart recited in claim 1, wherein the force sensor detects the translational force and the rotational force based on a potential difference across an electrical conductor that is transverse to an electric current in the electrical conductor and to an applied magnetic field perpendicular to the electric current.

10. The autonomous robotic cart recited in claim 1, wherein the holonomic, omnidirectional, and backdriveable mechanical drive unit includes four powered wheels.

11. The autonomous robotic cart recited in claim 1, wherein the holonomic, omnidirectional, and backdriveable mechanical drive unit includes four unpowered wheels.

12. The autonomous robotic cart recited in claim 1, wherein the holonomic, omnidirectional, and backdriveable mechanical drive unit includes two powered bidirectional wheels coupled with a turntable rotating about an axis.

13. The autonomous robotic cart recited in claim 12, wherein an unpowered freely rotating caster wheel is coupled with the turntable.

14. The autonomous robotic cart recited in claim 1, wherein the autonomous robotic cart includes a light component configured to illuminate a region of the one or more shelves corresponding to a current or intended location of an object designated for placement or retrieval.

15. The autonomous robotic cart recited in claim 1, wherein the autonomous robotic cart includes a light component configured to illuminate a region of the physical environment corresponding to a current or intended location of an object designated for placement or retrieval.

16. The autonomous robotic cart recited in claim 1, wherein autonomously navigating the physical environment involves moving along a nominal trajectory that includes one or more waypoints while avoiding objects encountered along the nominal trajectory.

17. A method of controlling an autonomous robotic cart that includes a chassis including one or more shelves, a holonomic, omnidirectional, and backdriveable mechanical drive unit coupled with the chassis, and a plurality of sensors coupled with the chassis, the plurality of sensors including a plurality of visible light cameras, the method comprising: transmitting a first one or more instructions to the holonomic, omnidirectional, and backdriveable mechanical drive unit to autonomously navigate a physical environment to execute one or more navigation goals determined based on communication with a remote computing system configured to manage a fleet of robots including the autonomous robotic cart; detecting a translational force and a rotational force exerted on a handlebar included in a handlebar unit coupled with the chassis, the handlebar being vertically positioned orthogonal to the planar surface; determining an output force vector based on the translational force, the rotational force, sensor data received from the plurality of sensors, and a force multiplier applied to the translational force and the rotational force such that a translational force of interaction and a rotational force of interaction exerted by the handlebar falls below a designated threshold; and transmitting a second one or more instructions to the holonomic, omnidirectional, and backdriveable mechanical drive unit to cause the autonomous robotic cart to move translationally and rotationally in a direction corresponding to the output force vector.

18. One or more non-transitory computer readable media having instructions stored thereon for performing a method of controlling an autonomous robotic cart that includes a chassis including one or more shelves, a holonomic and omnidirectional mechanical drive unit coupled with the chassis, and a plurality of sensors coupled with the chassis, the plurality of sensors including a plurality of visible light cameras, the method comprising: transmitting a first one or more instructions to the holonomic, omnidirectional, and backdriveable mechanical drive unit to autonomously navigate a physical environment to execute one or more navigation goals determined based on communication with a remote computing system configured to manage a fleet of robots including the autonomous robotic cart; detecting a translational force and a rotational force exerted on a handlebar included in a handlebar unit coupled with the chassis, the handlebar being vertically positioned orthogonal to the planar surface; determining an output force vector based on the

translational force, the rotational force, sensor data received from the plurality of sensors, and a force multiplier applied to the translational force and the rotational force such that a translational force of interaction and a rotational force of interaction exerted by the handlebar falls below a designated threshold; and transmitting a second one or more instructions to the holonomic, omnidirectional, and backdriveable mechanical drive unit to cause the autonomous robotic cart to move translationally and rotationally in a direction corresponding to the output force vector.
